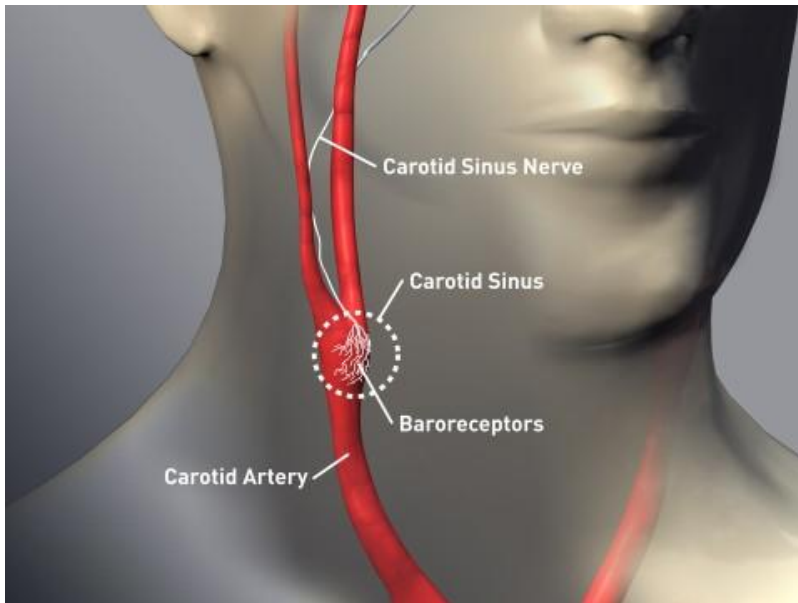


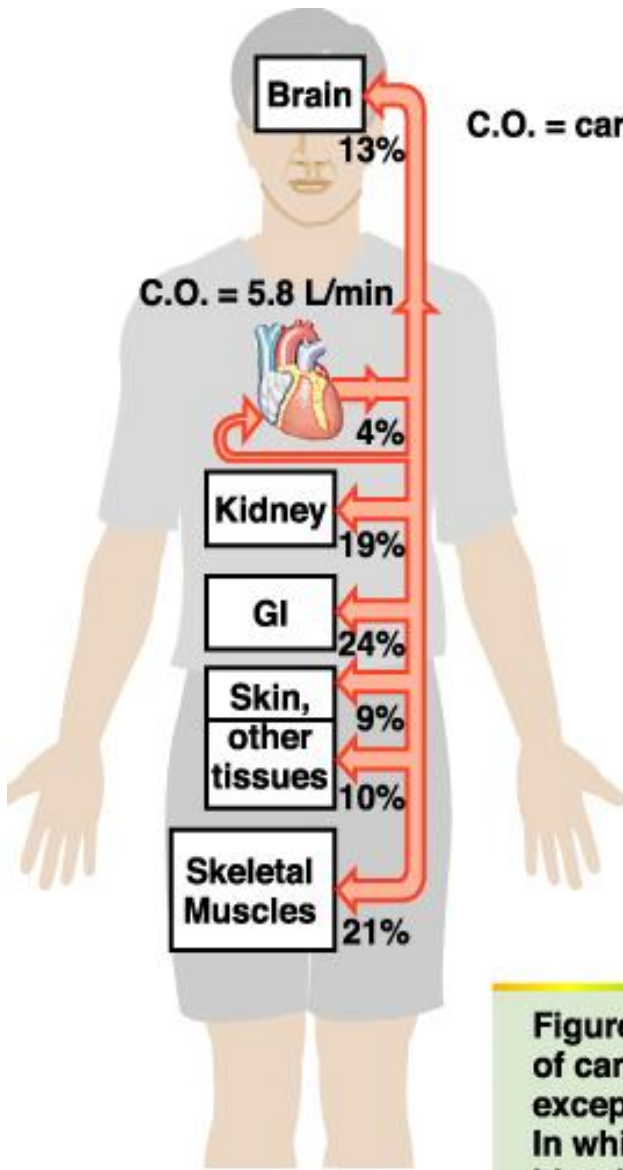
Section 4 Cardiovascular Regulatory Mechanisms



马泽刚

博雅楼507

mazegang2000@163.com



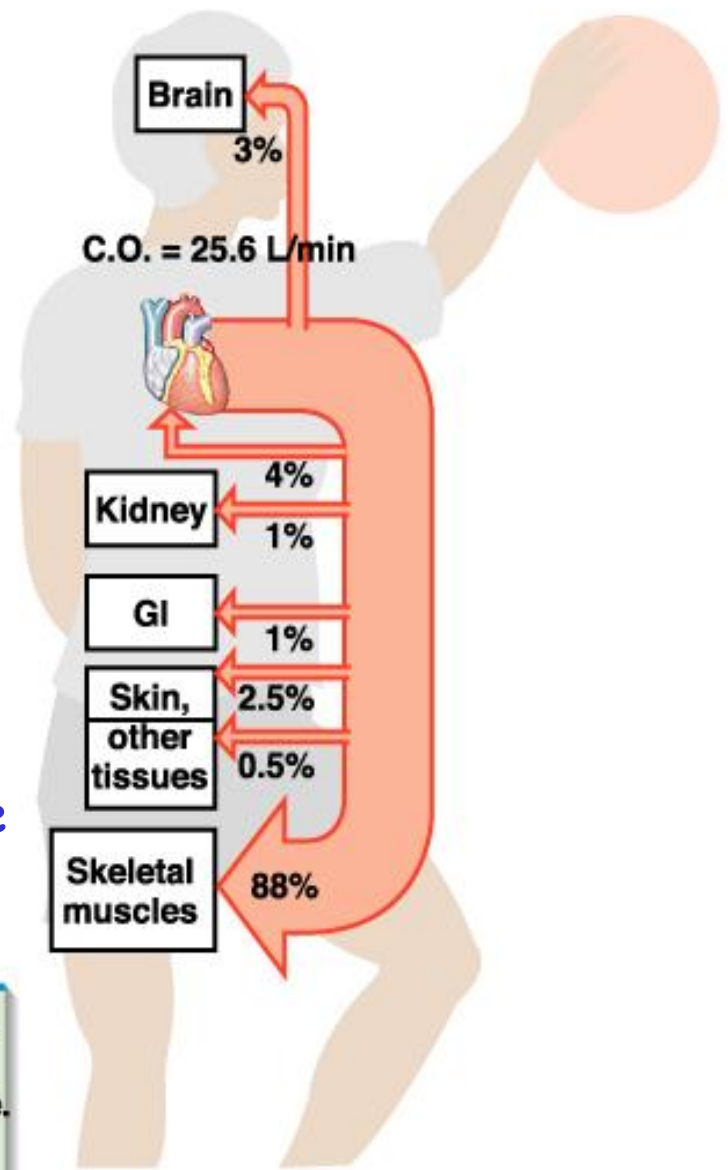
At rest

C.O. = cardiac output

➤ Cardiovascular regulatory mechanisms increase the blood supply to **active tissues**.

➤ In the face of challenges, they maintain the blood flow to the vital organs like **heart and brain**.

Figure question: The percentage of cardiac output to all tissues except muscle falls with exercise. In which tissues does actual blood flow decrease?



Vigorous exercise

Section outline

•Neural regulation

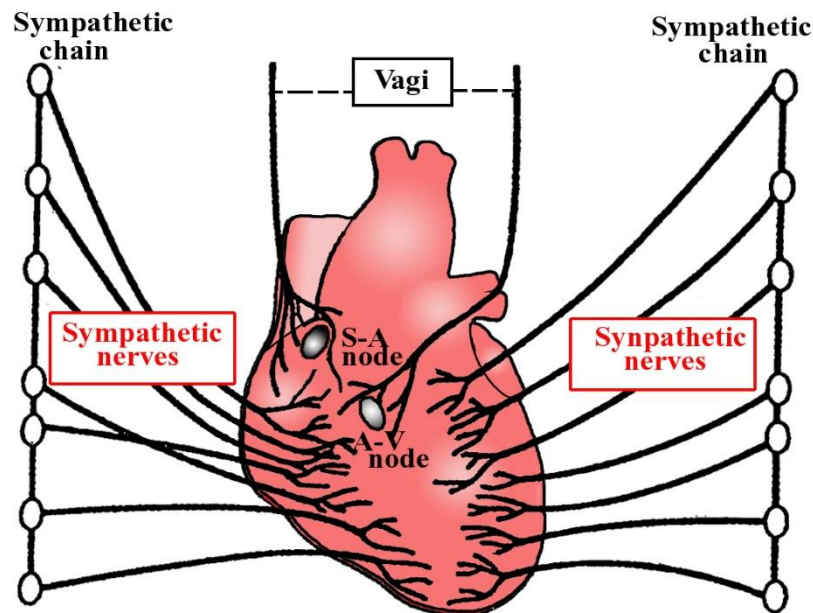
- Innervations of heart and blood vessels
- Cardiovascular center
- Cardiovascular reflex

•Humoral regulation

- Renin-angiotensin system (RAS)
- Epinephrine & Norepinephrine (E & NE)
- Vasopressin (VP)
- Atrial natriuretic peptide (ANP)

•Autoregulation

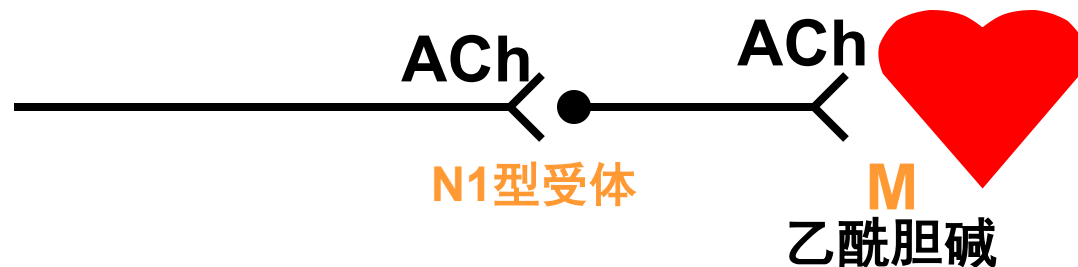
Cardiac Innervation



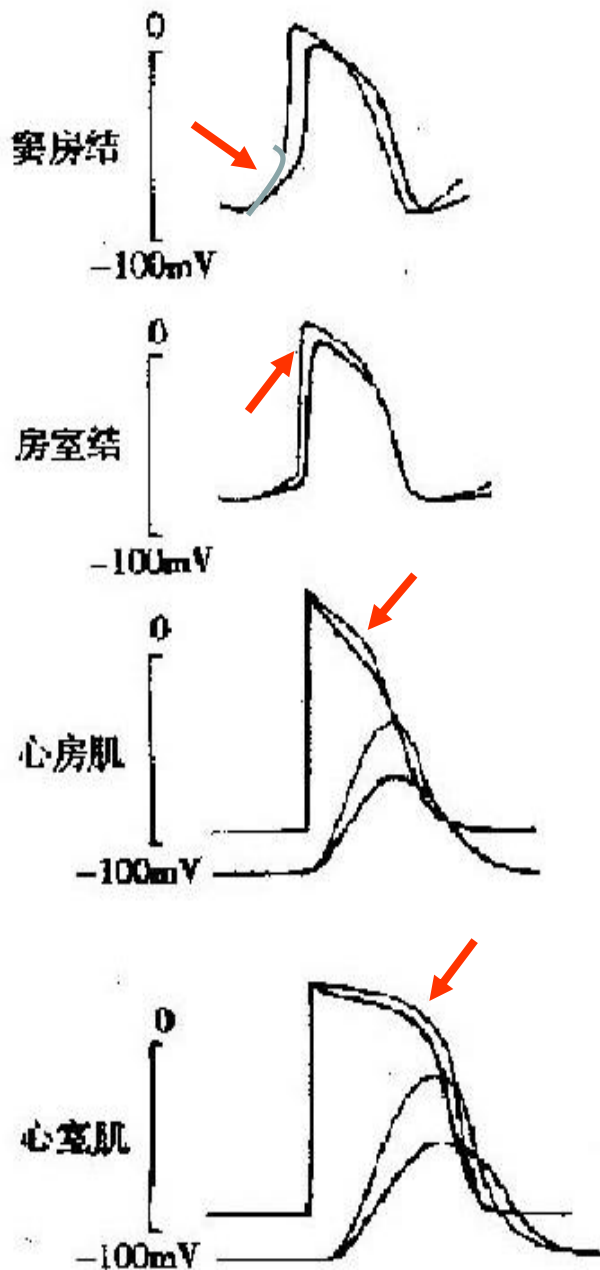
— 心交感神经



— 心迷走神经



交感神经作用



NE — G蛋白—AC—cAMP↑-PKA
— Ca^{2+} 通道磷酸化而激活— Ca^{2+} 内流↑

• 正性变时作用 — 心率增快

∴ $I_{\text{Ca-T}}$ ↑ I_f ↑ — 4期自动去极化速度↑

• 正性变传导作用 — 传导加快

∴ $I_{\text{Ca-L}}$ ↑ — 0期除极的速度和幅度↑

• 正性变力作用 — 收缩增强

∴ (1) $I_{\text{Ca-L}}$ ↑, 2期 Ca^{2+} 内流↑

(2) 钙触发钙释放, 肌浆网释放 Ca^{2+} ↑

(3) 受磷蛋白(PLB)磷酸化, 钙泵回收 Ca^{2+} ↑

(4) 正性变传导使收缩更趋于同步化



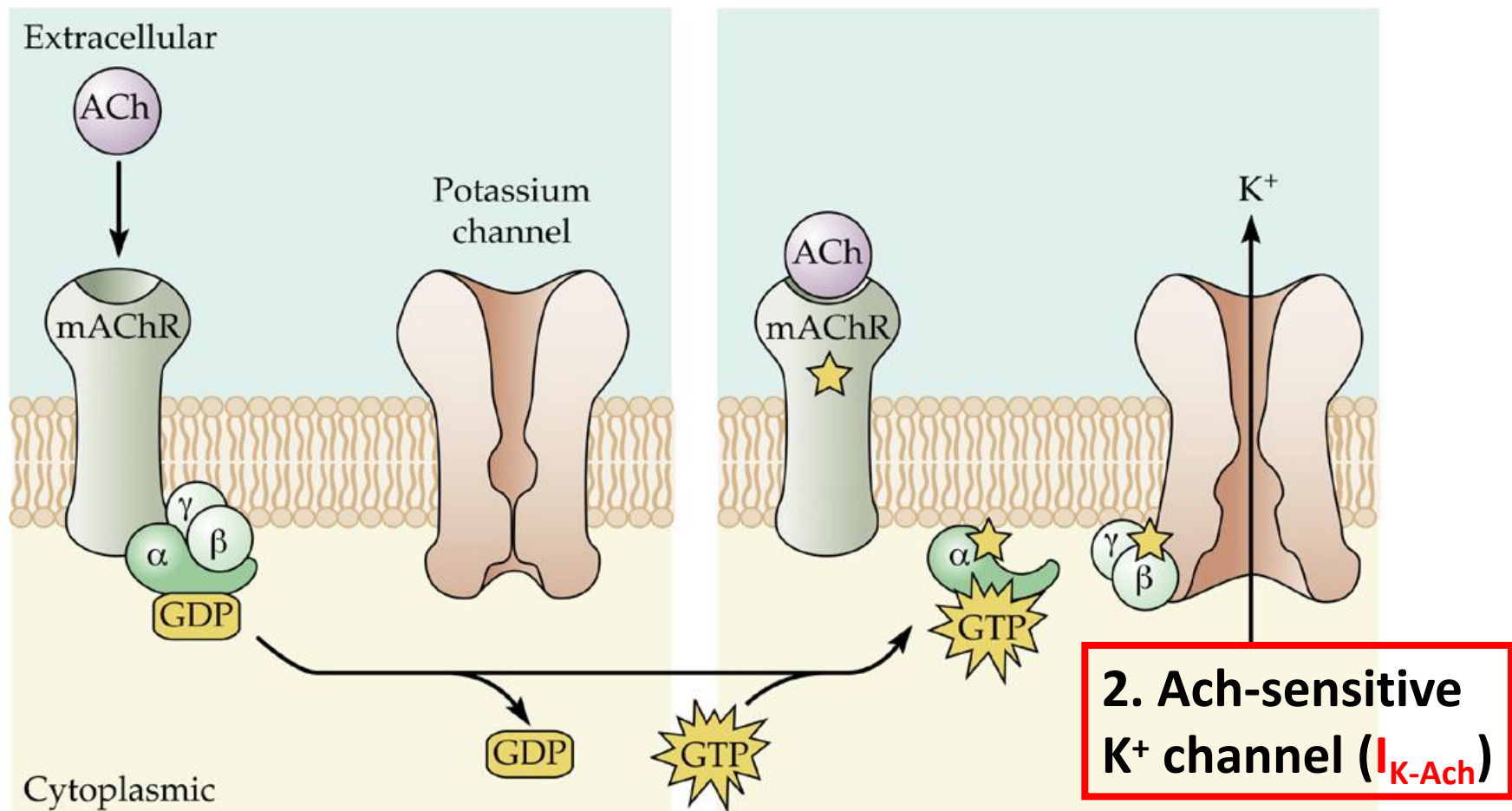
Sir James Black with his Nobel prize medal, awarded to him for his work on drug development, namely his invention of the **beta-blocker drug propranolol** (普萘洛尔/心得安, 1964).

NOBEL prize-winning scientist (1988) Sir James Black has died at the age of 85 (March 23, 2010)

β 1 blocker: metoprolol 美托洛尔

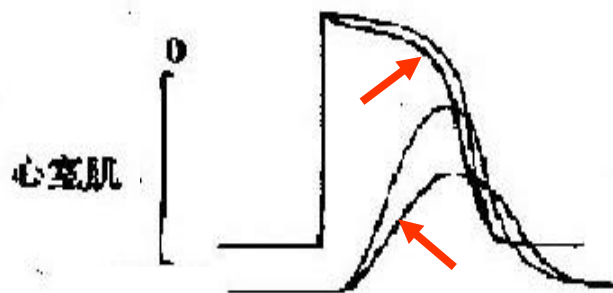
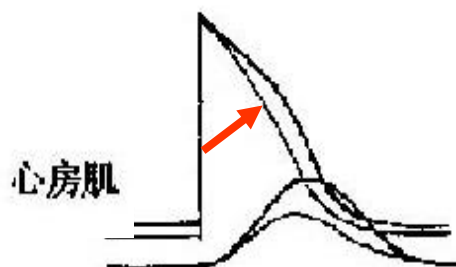
Acetylcholine

1. G蛋白-AC-cAMP ↓ 与去甲肾上腺素相反的效应



存在于心房肌和窦房结

副交感神经作用



ACh — G蛋白 — cAMP↓ — Ca^{2+} 内流↓
 K^{+} 外流↑

- 负性变时作用 — 心率减慢

- ∴ (1) $I_{\text{Ca-T}}$ ↓ I_{f} ↓ — 4期自动去极化速度↓
- (2) K^{+} ↑, 4期最大复极电位更负

- 负性变传导作用 — 传导减慢

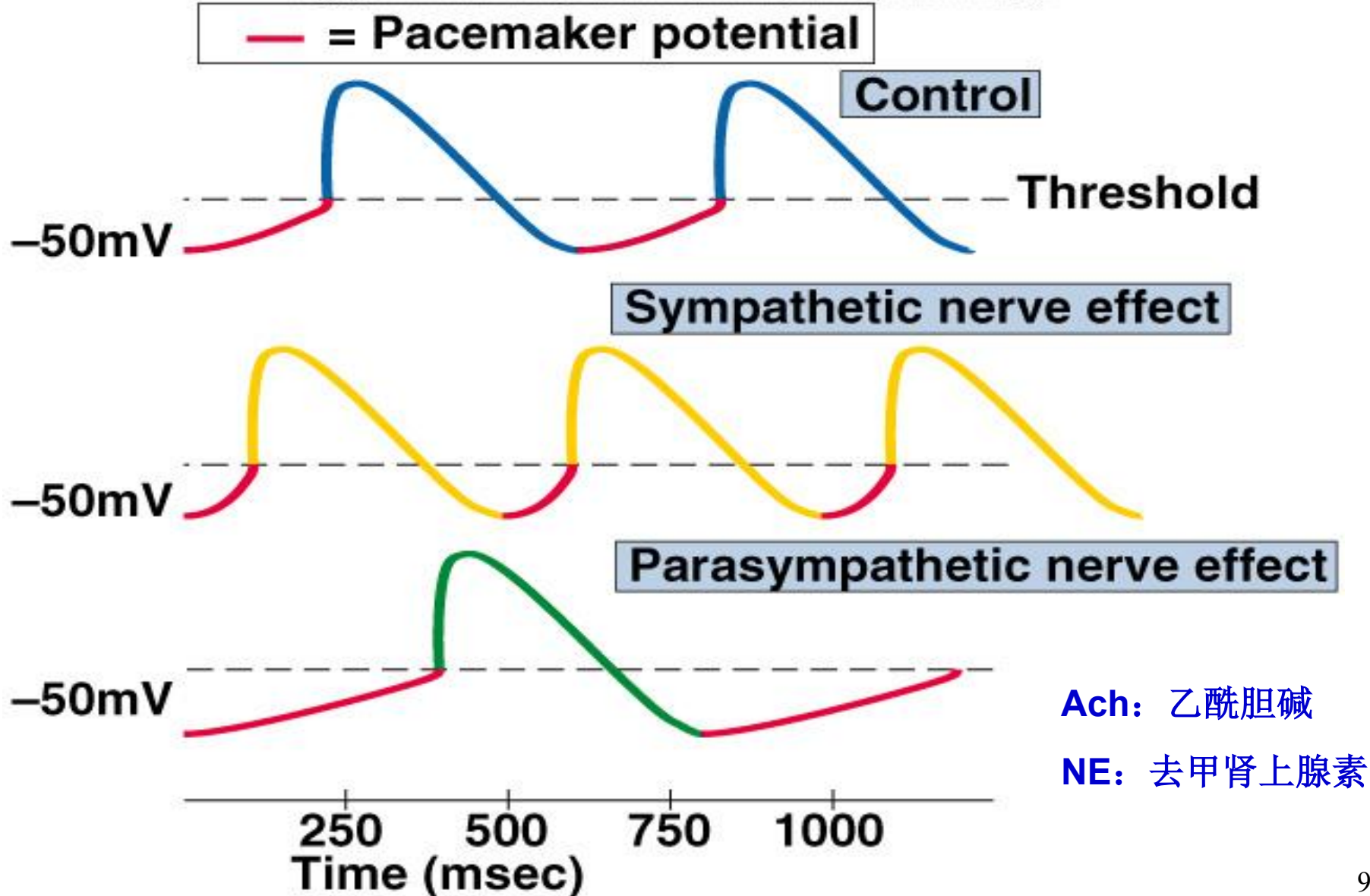
- ∴ Ca^{2+} ↓ — 0期除极的速度和幅度↓

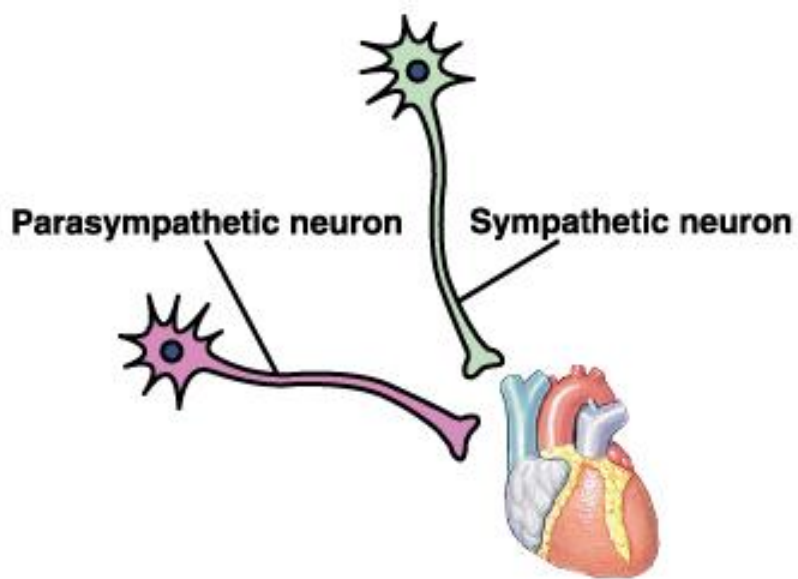
- 负性变力作用 — 收缩减弱

- ∴ K^{+} ↑
 Ca^{2+} ↓ — 平台期缩短 — Ca^{2+} ↓

Modulation Pacemaker Activity.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

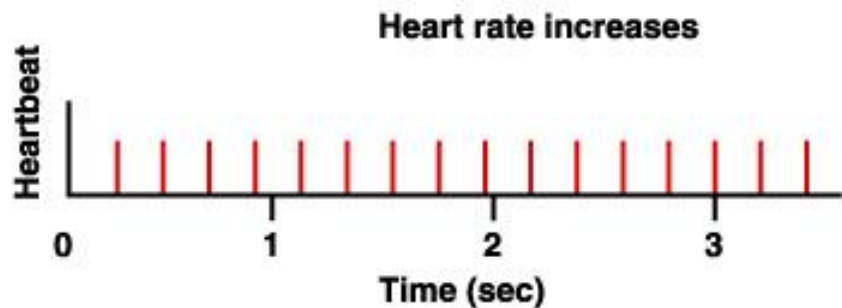
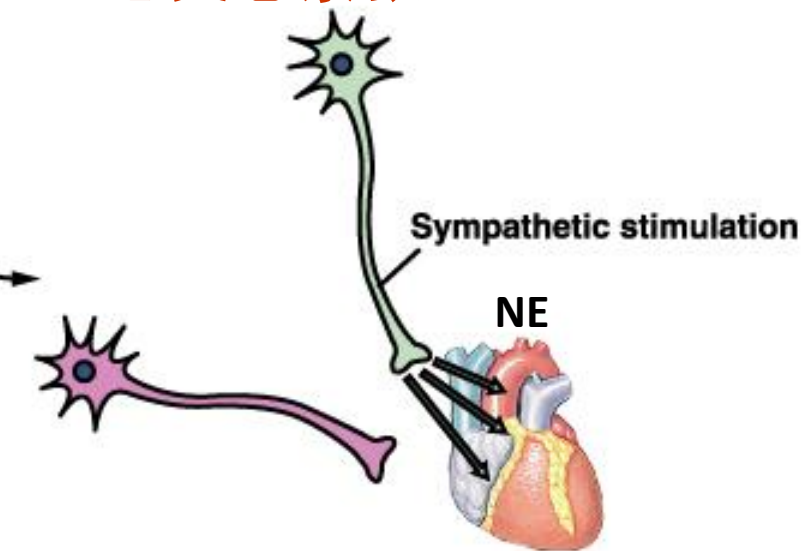


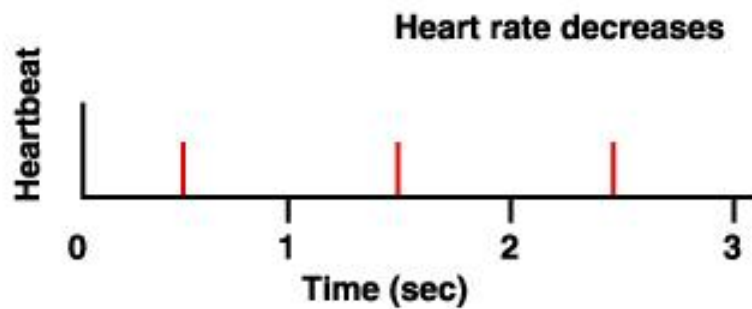
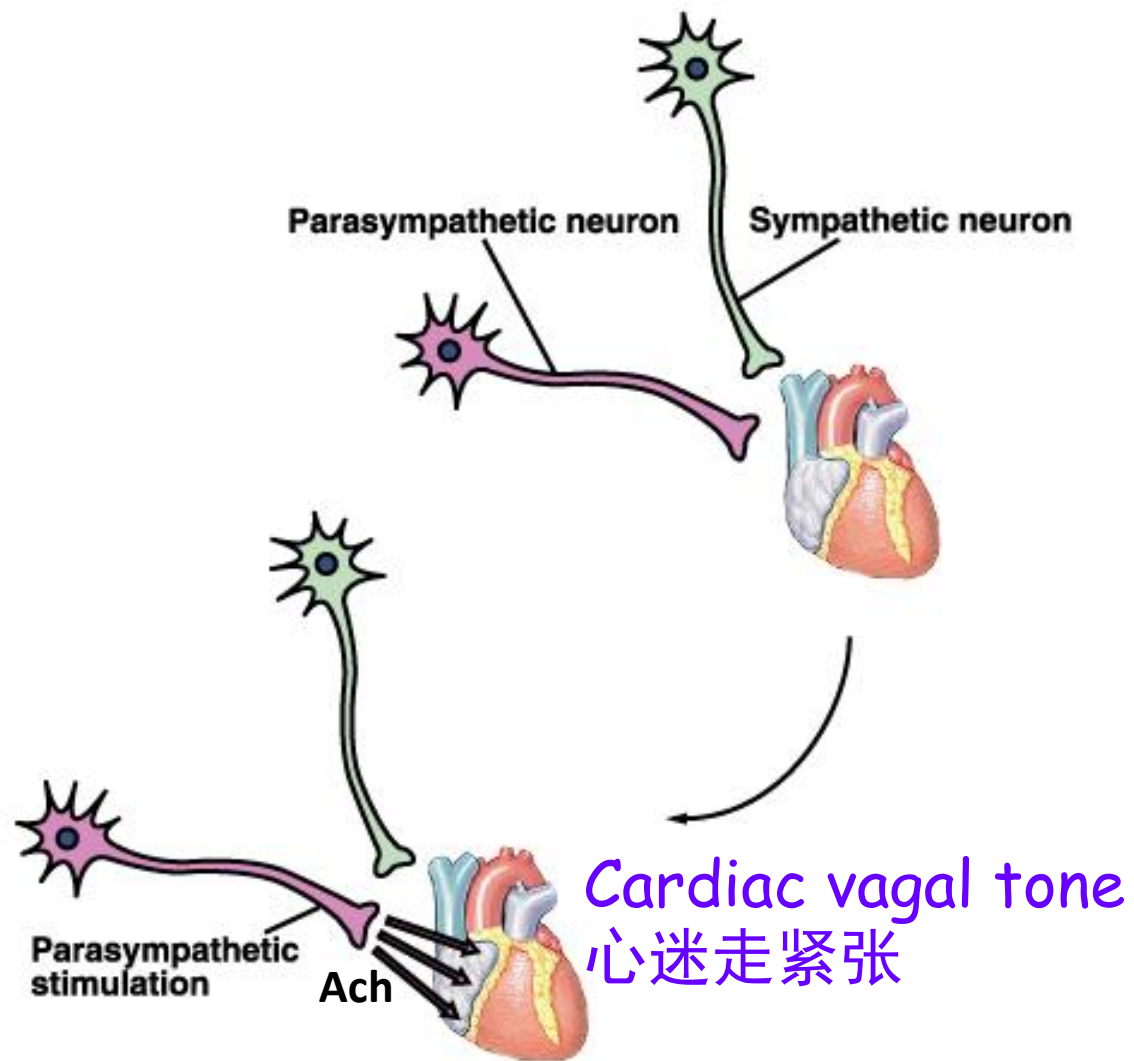


Cardiac sympathetic tone
心交感紧张

Tonus

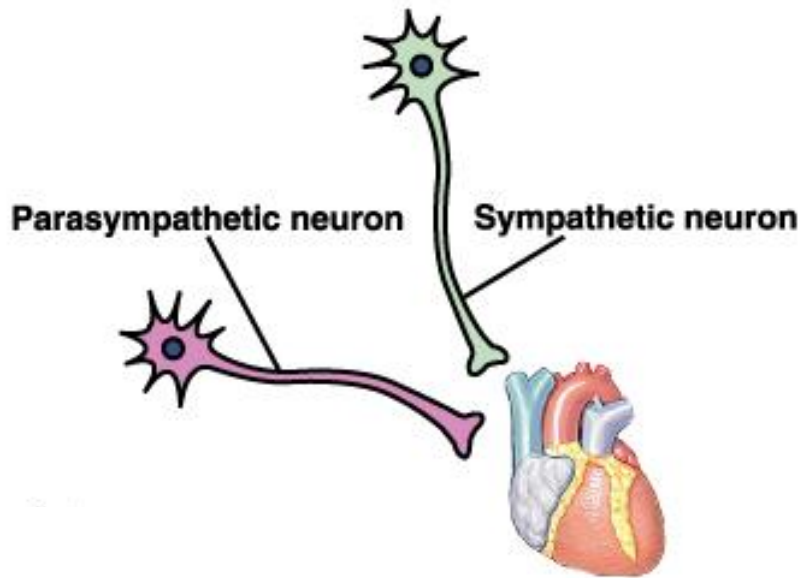
神经或肌肉等组织保持一定程度的持续活动状态





心交感紧张 vs 心迷走紧张

- 相互拮抗
- 迷走优势



In humans in whom both norepinephrinergic and cholinergic systems are blocked, the heart rate is approximately 100, whose normal resting value is 70.

→ more amount of tonic vagal than sympathetic discharge at rest in humans.

Innervations of the Blood Vessels

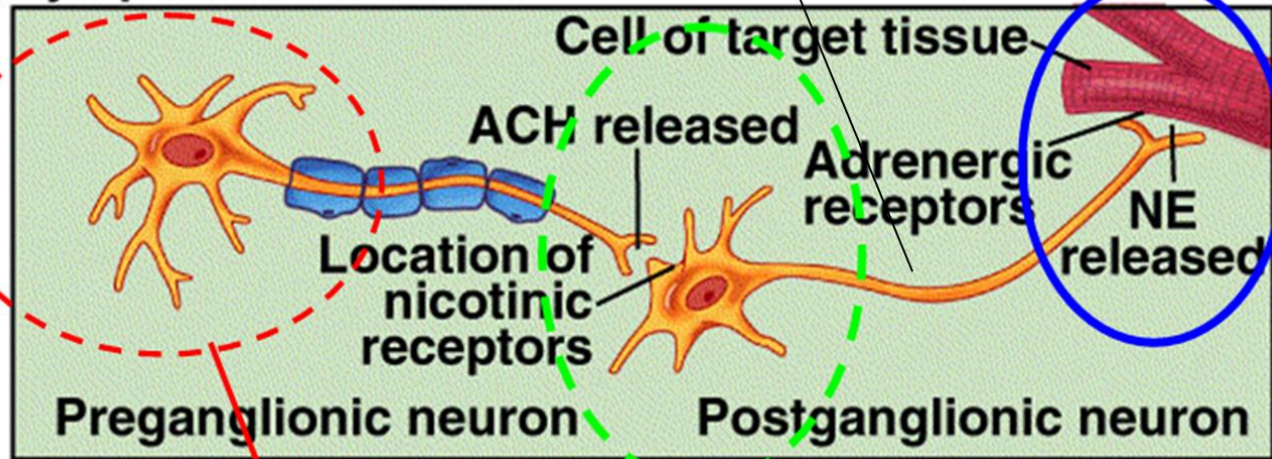
支配血管平滑肌（收缩和舒张活动）的神经称为血管运动神经（vasomotor nerve）

- 交感缩血管神经纤维(Vasoconstrictor fiber)
- 舒血管神经纤维 (Vasodilator fiber)
 - 交感舒血管
 - 副交感舒血管
 - 脊髓后根舒血管纤维
 - 肽类舒血管神经纤维

Sympathetic vasoconstrictor fibers

Focus on this synapse

Sympathetic division

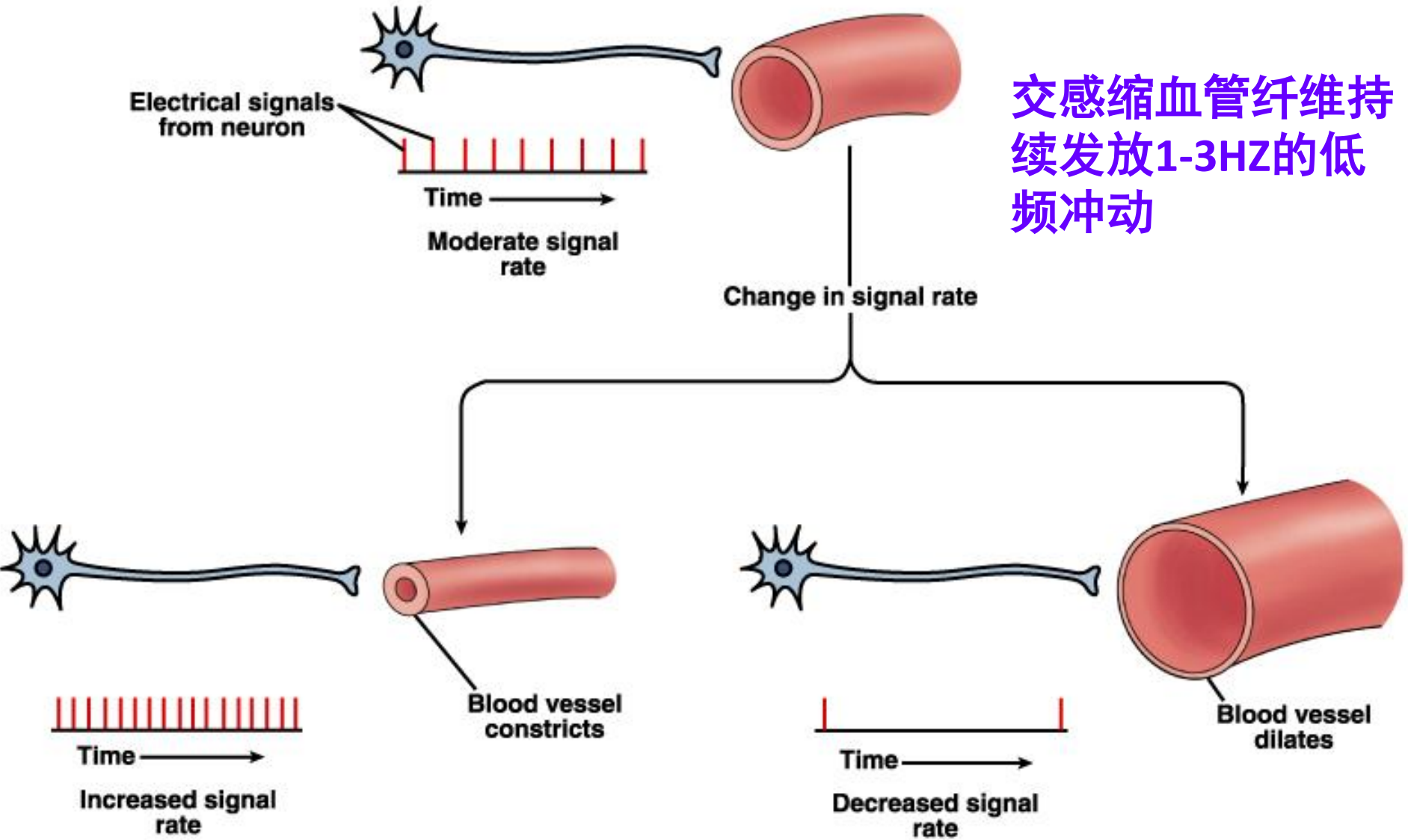


α -receptor
→ 血管收缩

Cell body in spinal cord

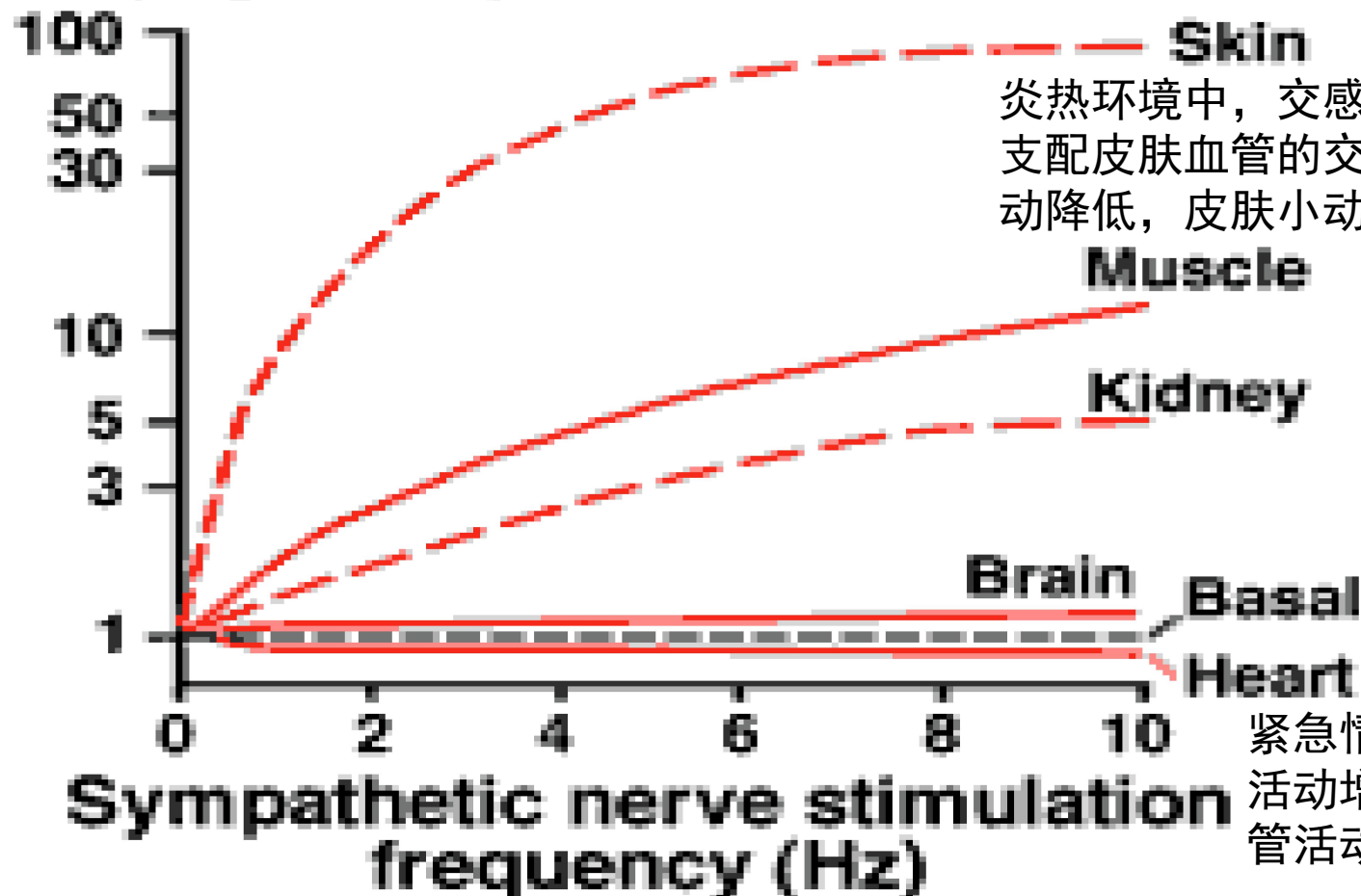
autonomic ganglion

Sympathetic vasoconstrictor tone 交感缩血管紧张



Sympathetic vasoconstrictor fibers in various organs

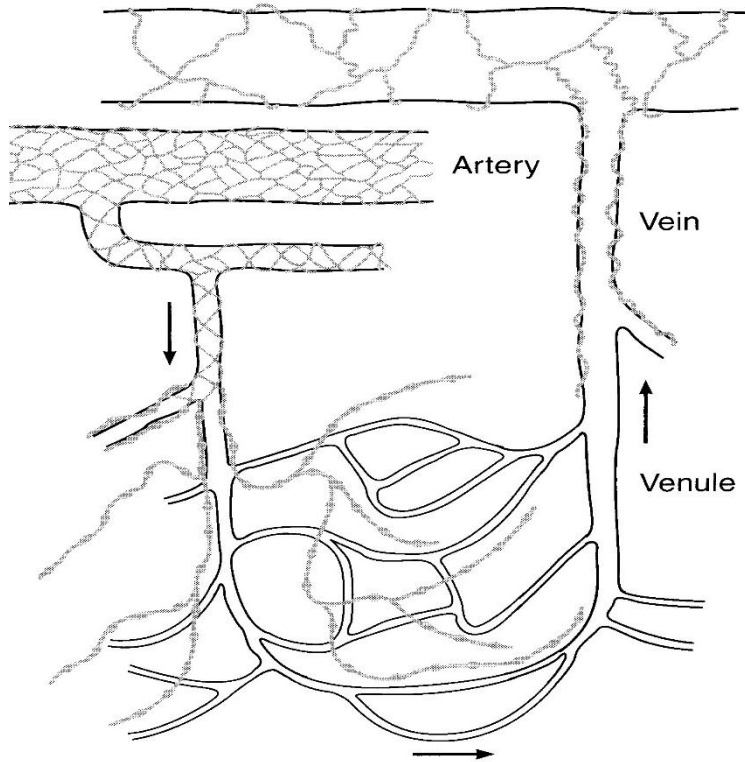
Vascular resistance
(log scale)



炎热环境中，交感神经活动减弱，支配皮肤血管的交感神经紧张性活动降低，皮肤小动脉舒张。

紧急情况时，交感神经活动增强，但对心脑血管活动影响较小。

Sympathetic vasoconstrictor fibers in one organ



- 动脉密度高于静脉
- 微动脉中密度最高
- 毛细血管前括约肌中密度最低
- 毛细血管不受支配

	交感缩血管	交感舒血管	副交感舒血管
末梢递质	去甲肾上腺素	乙酰胆碱	乙酰胆碱
受体	α 、 β_2	M受体	M受体
效应器	绝大多数血管	骨骼肌血管	少数器官（如消化、外生殖器）
效应	α - 血管收缩(主要) β_2 - 血管舒张(弱)	血管舒张	血管舒张
紧张性活动	交感缩血管紧张	无	无
生理作用	调节外周阻力和血流量	与骨骼肌血流增加有关	调节局部血流

心血管中枢 (Cardiovascular center)

➤ Cardiovascular center:

中枢神经系统中与控制心血管活动有关的神经元集中的部位

➤ 脊髓，延髓，下丘脑，大脑及小脑等

脊髓：胸腰段灰质中间外侧柱(IML)中的神经元，是中枢调控心血管活动的最后传出通路。

Medullary(延髓) cardiovascular center

- 产生和维持心交感神经和交感缩血管神经紧张

延髓头端腹外侧区(RVLM)

- 产生和维持心迷走紧张

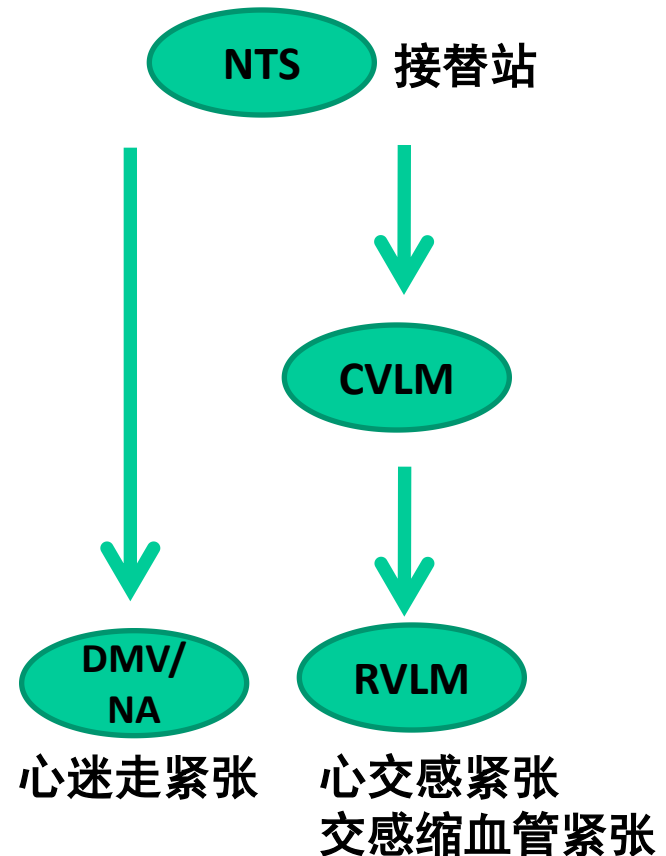
疑核(NA) & 迷走神经背核(DMV)

- 传入神经的接替站

孤束核(NTS)

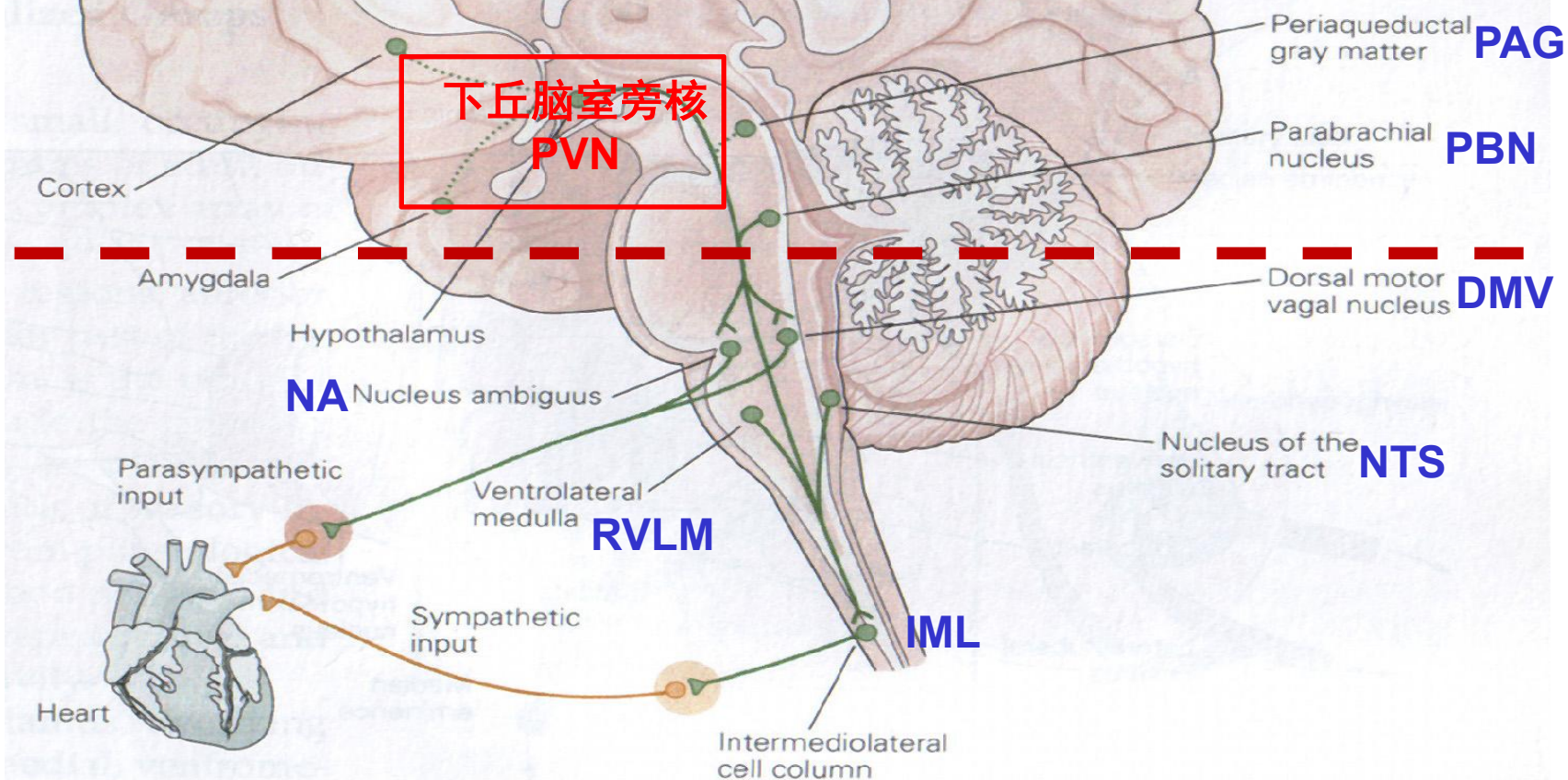
- 抑制RVLM

延髓尾端腹外侧区(CVLM)



Higher cardiovascular center

复杂整合



Cardiovascular reflex

- **Baroreceptor reflex (压力感受性反射)**
- **Chemoreceptor reflex (化学感受性反射)**
- **Cardiopulmonary receptor reflex
(心肺感受器引起的心血管反射)**

Baroreceptor Reflex

当动脉血压**突然**升高时，可反射性引起。。。 (心血管活动抑制)

反射弧

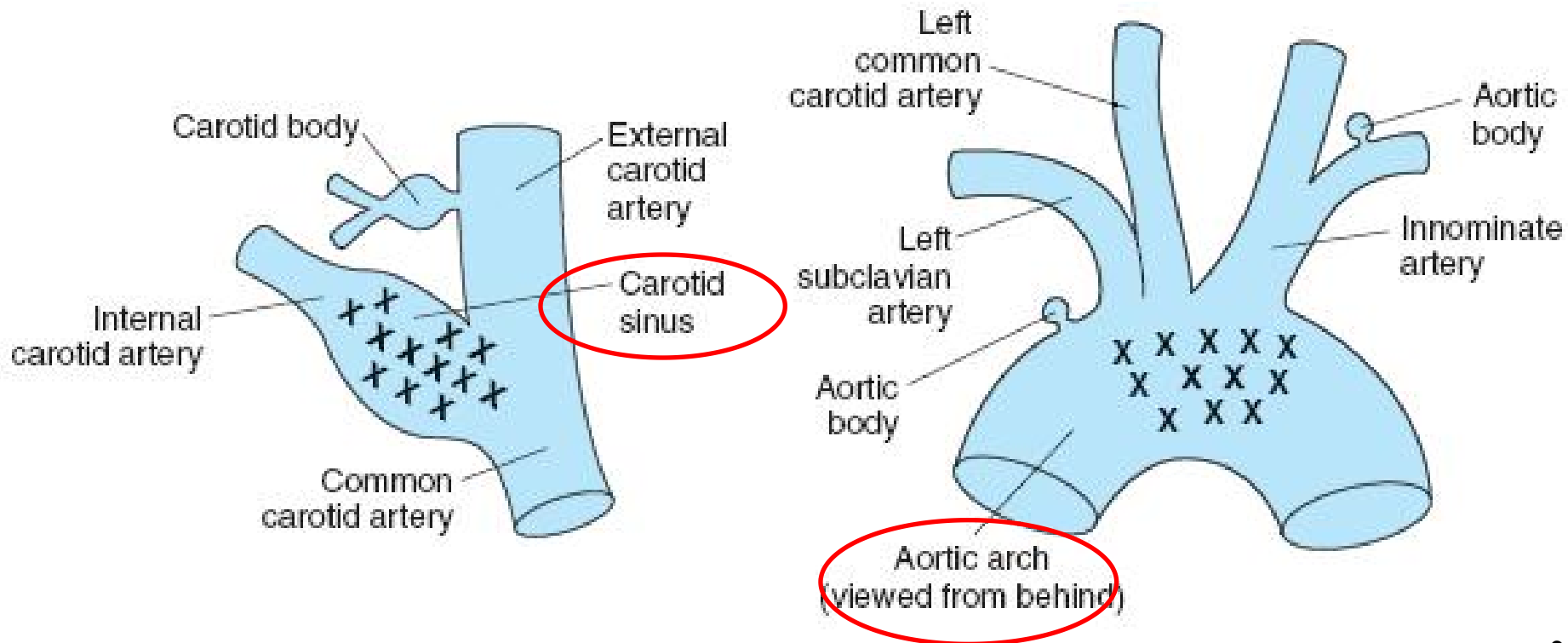
- 感受器 — 颈动脉窦(carotid sinus)
主动脉弓(aortic arch)
- 传入神经 — 窦神经(加入舌咽神经)、
迷走神经
- 中枢 — 延髓
- 传出神经 — 心迷走神经
心交感神经
交感缩血管神经
- 效应器 — 心脏和血管

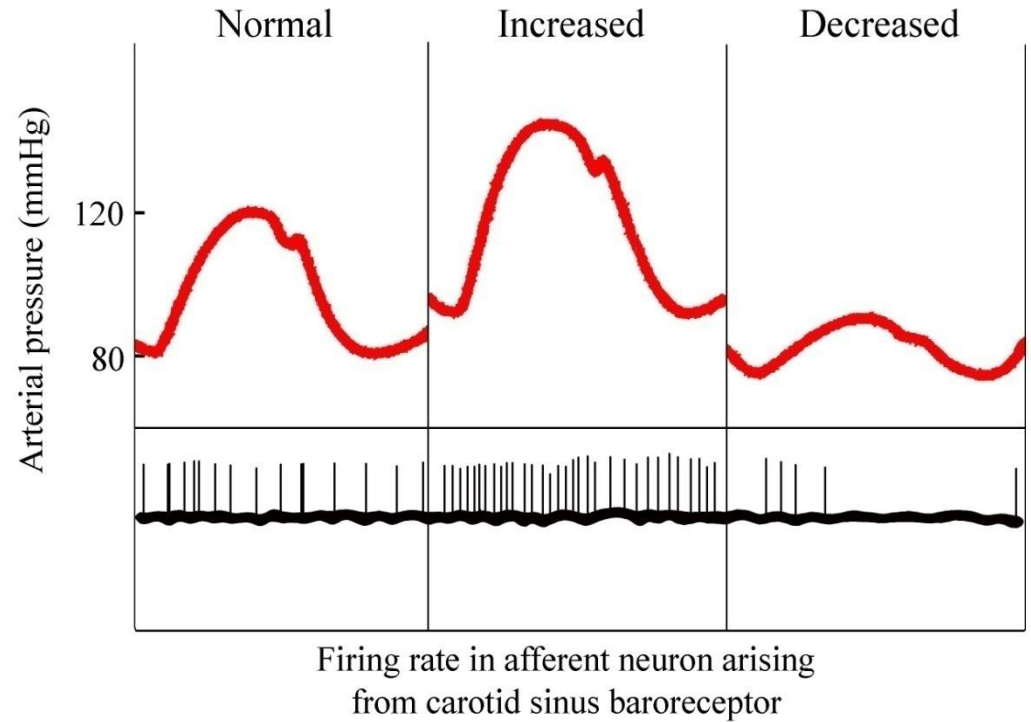
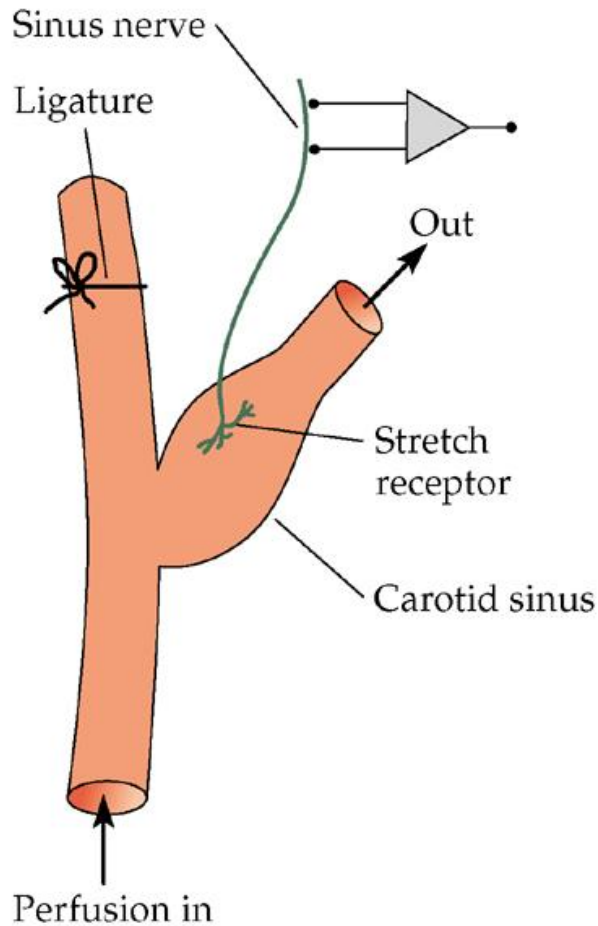


Baroreceptors Reflex

压力感受器(baroreceptor)

位置：颈动脉窦和主动脉弓血管外膜下的感觉神经末梢

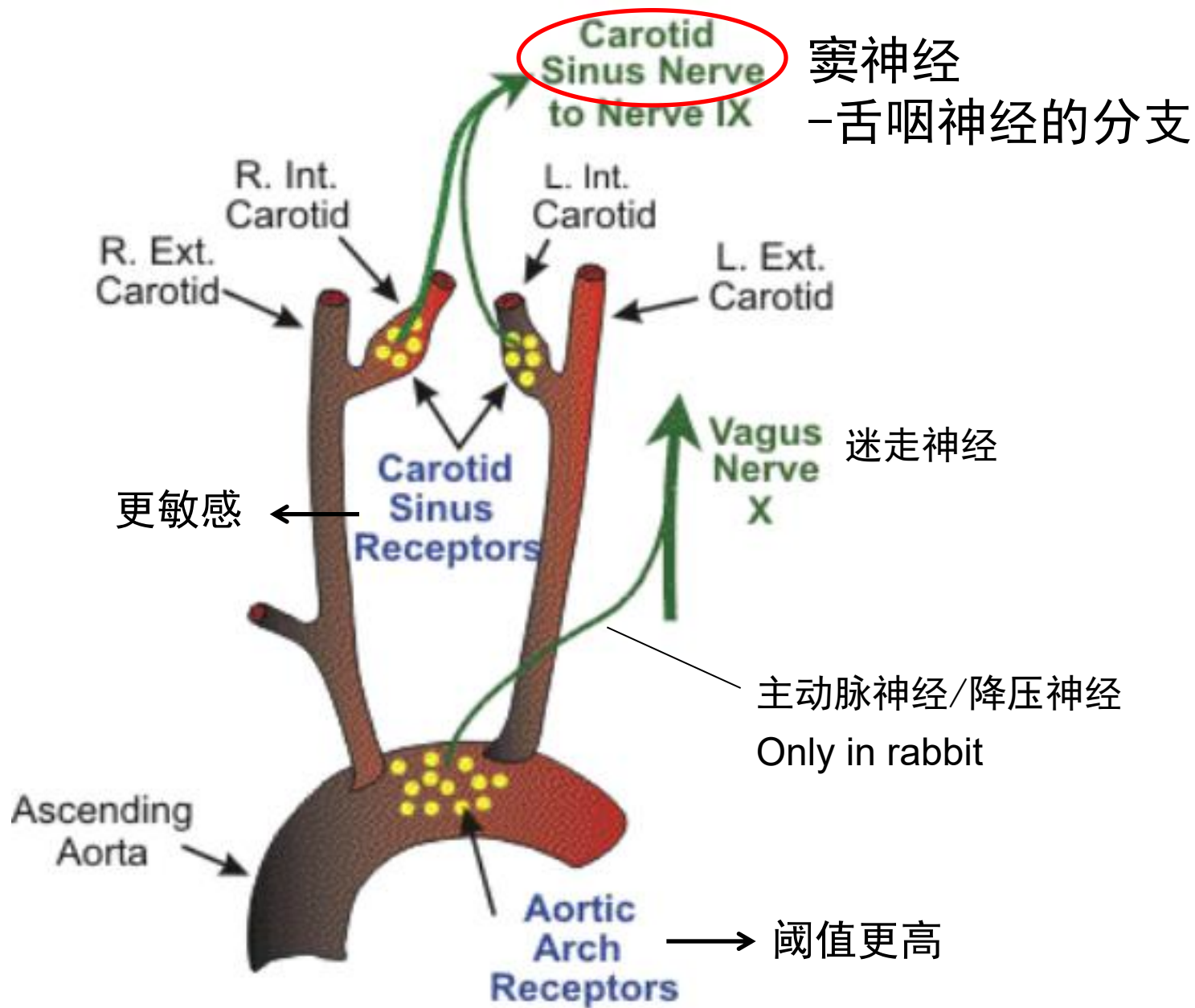


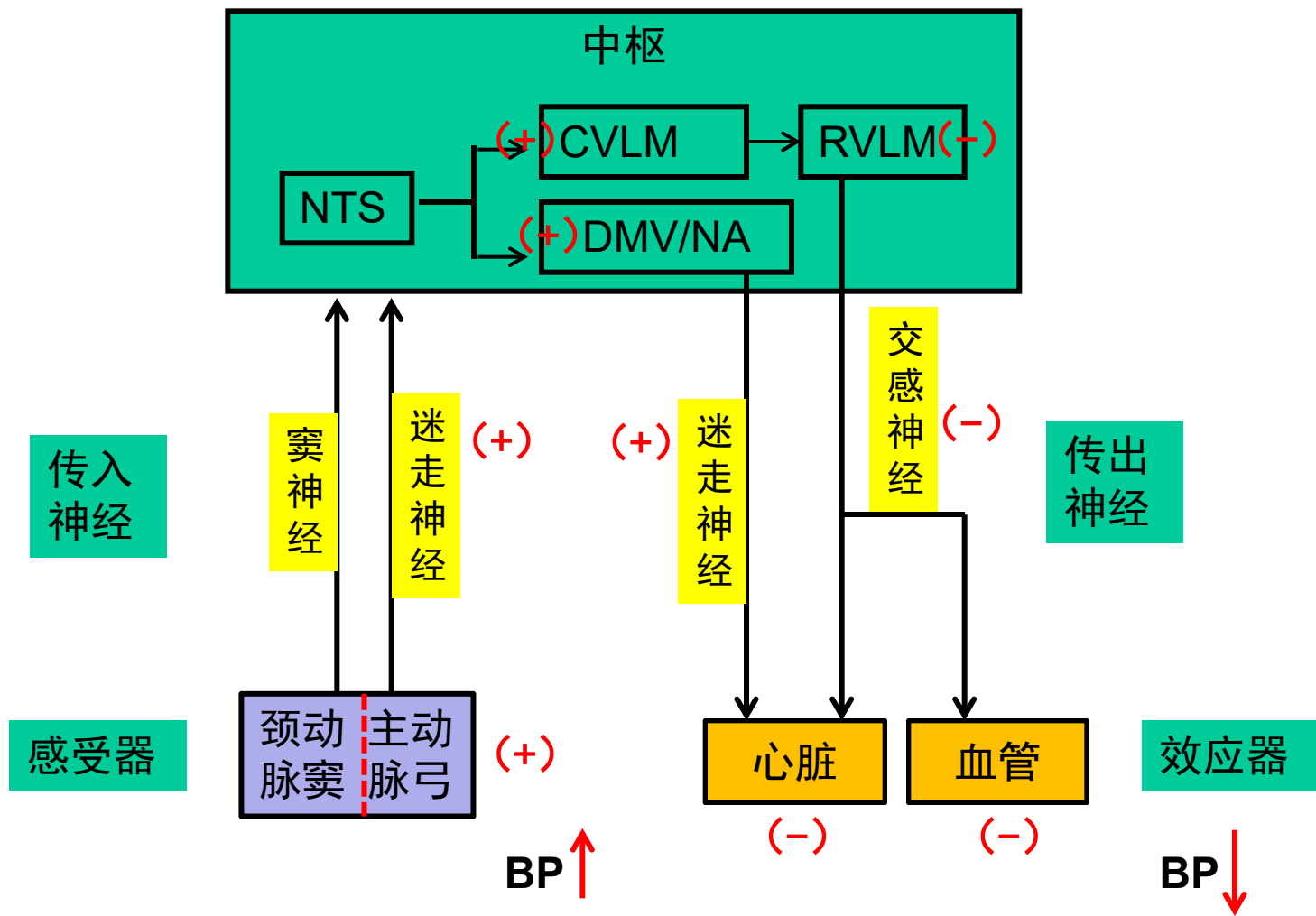


Baroreceptors are **stretch** but not pressure receptors!

Stimulated by distension of the structures in which they are located

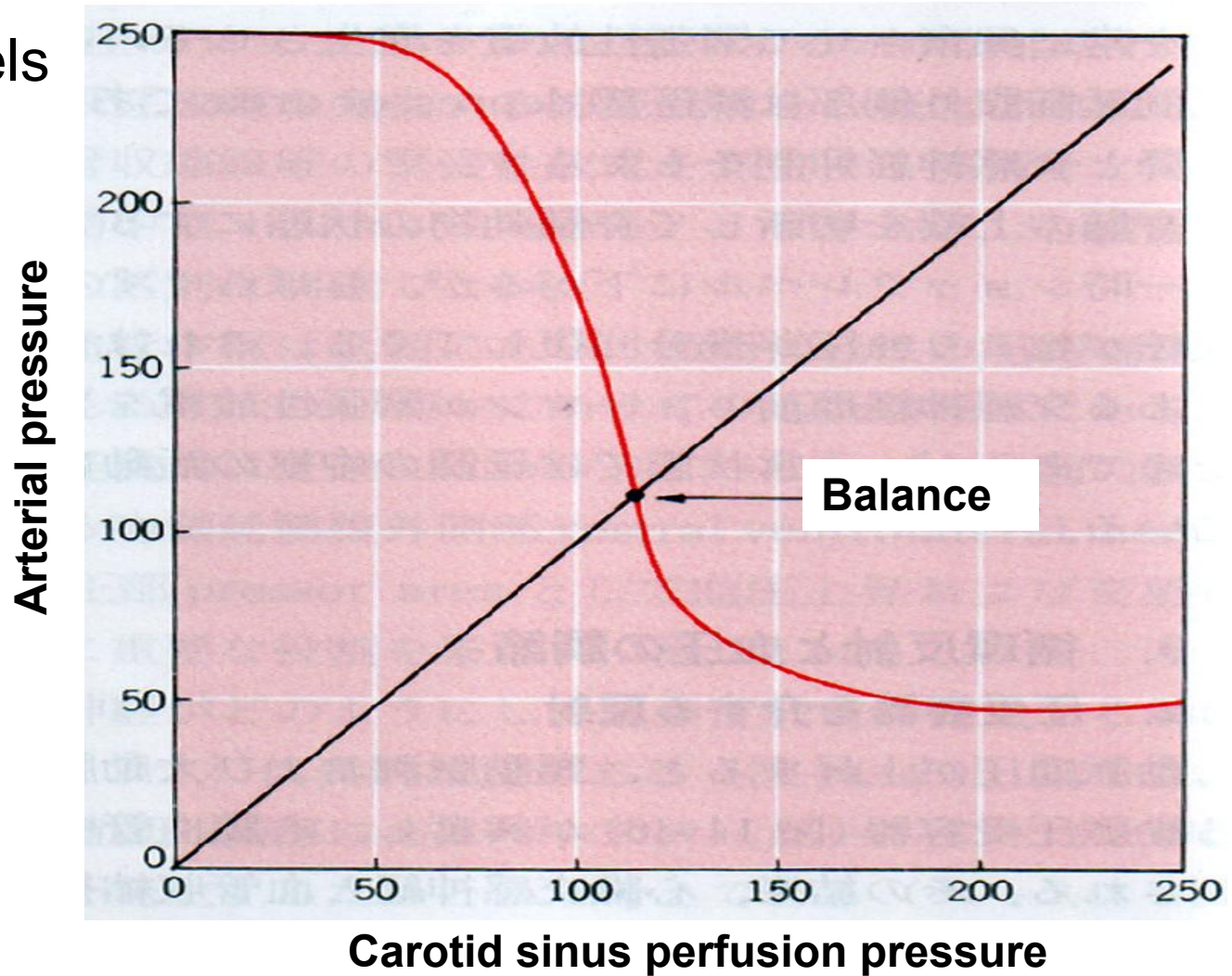
刺激：机械牵张



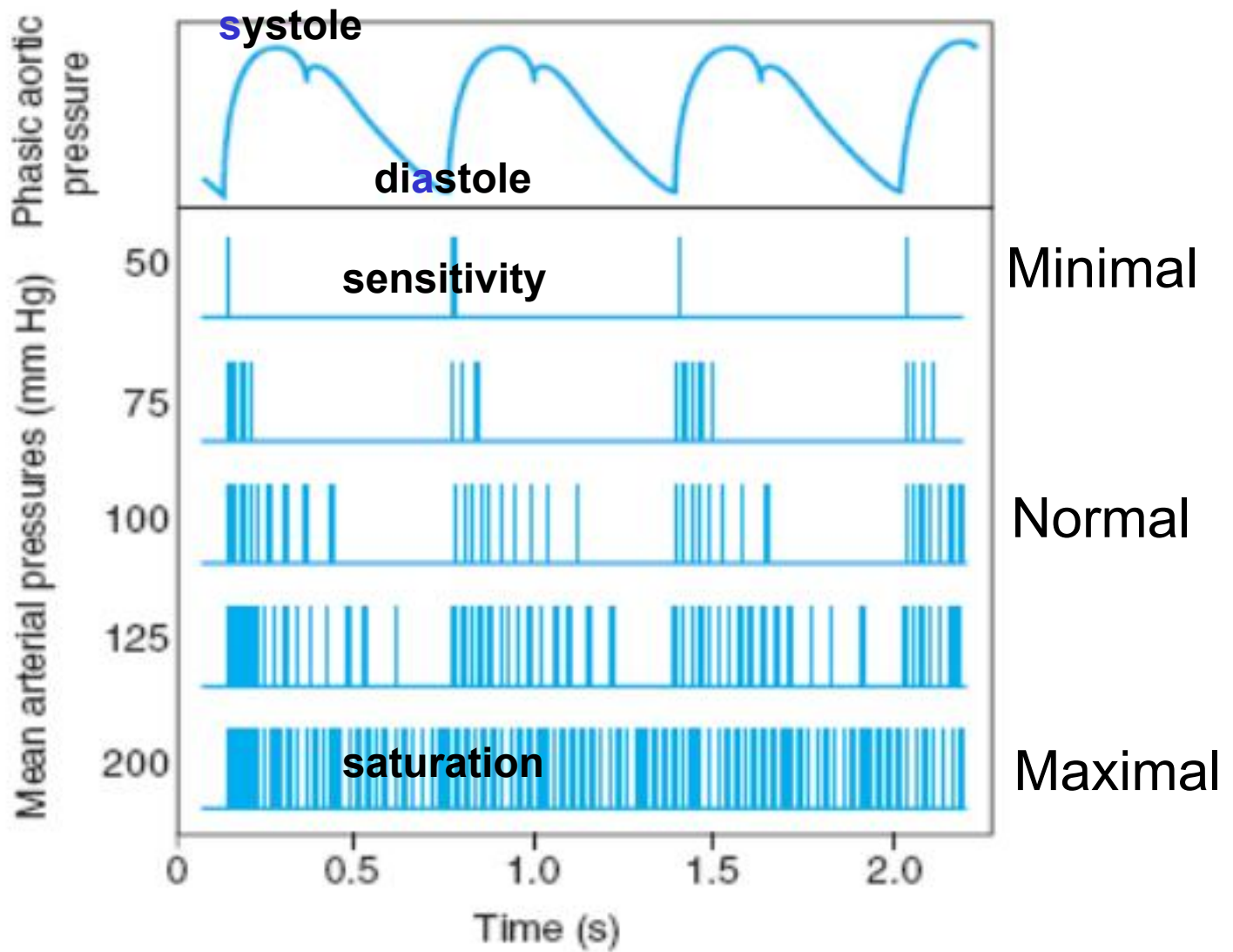


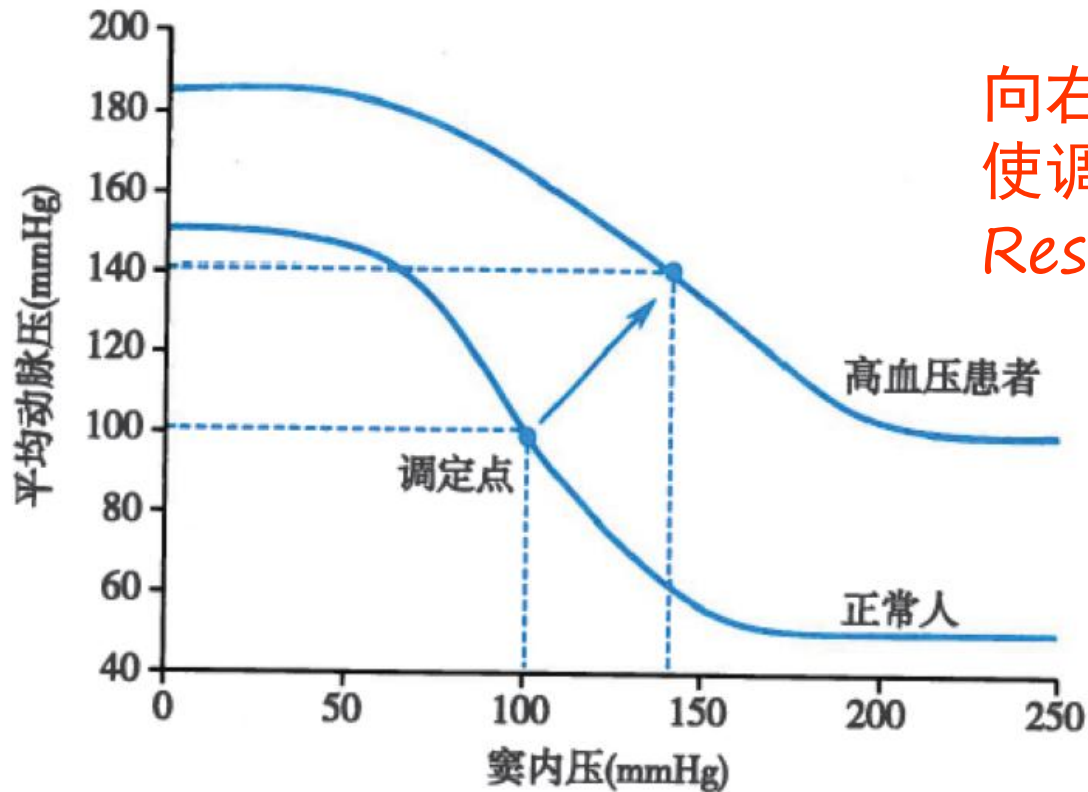
注意：双向效应！

Animal models



- When one carotid sinus is isolated and perfused, the relation between the perfusion pressure and the arterial pressure is essentially linear at perfusion pressures of 80–120 mm Hg
- There is no discharge and no regulation with the perfusion pressure is below 50 mm Hg.
- At perfusion pressures above 200 mm Hg, there is no further increase in response.





向右上方移位，
使调定点升高，
Resetting（重调定）

图 4-33 正常人和高血压患者的压力感受性反射功能曲线

在高血压的情况下，压力感受性反射功能曲线的工作范围发生改变，在高血压水平上保持血压相对稳定，但不能依赖压力感受性反射使高血压恢复正常。

思考：

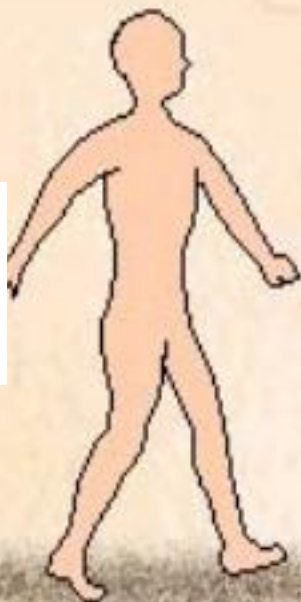
从卧位到站位，人体的血压有何变化，为什么？

卧位 → 立位

静脉回心血量↓，心输出量↓，
血压↓，压力感受器↓，血压↑

立位 → 卧位

静脉回心血量↑，心输出量↑，
血压↑，压力感受器↑，血压↓



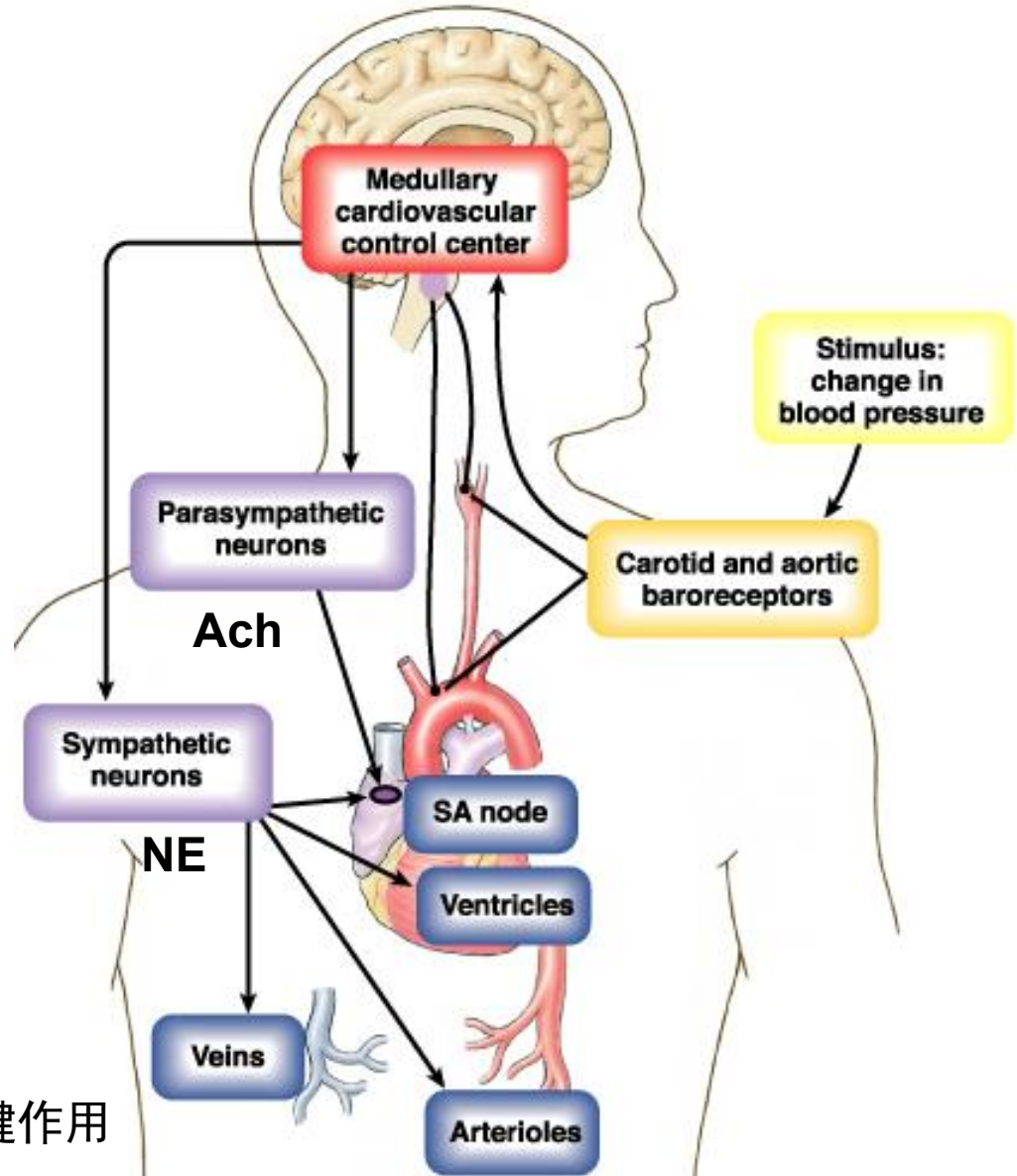
敏感性：快速变化 > 缓慢变化

Significance

✓ At the **very beginning** of systemic circulation, the receptors sense the small **fluctuations (波动)** of arterial pressure, thus constitute a reflex feedback mechanism that operate to **stabilize** the blood pressure and heart rate.

✓ Do not participate long-term AP regulation

在动脉血压的长期调节中不起关键作用



实验：

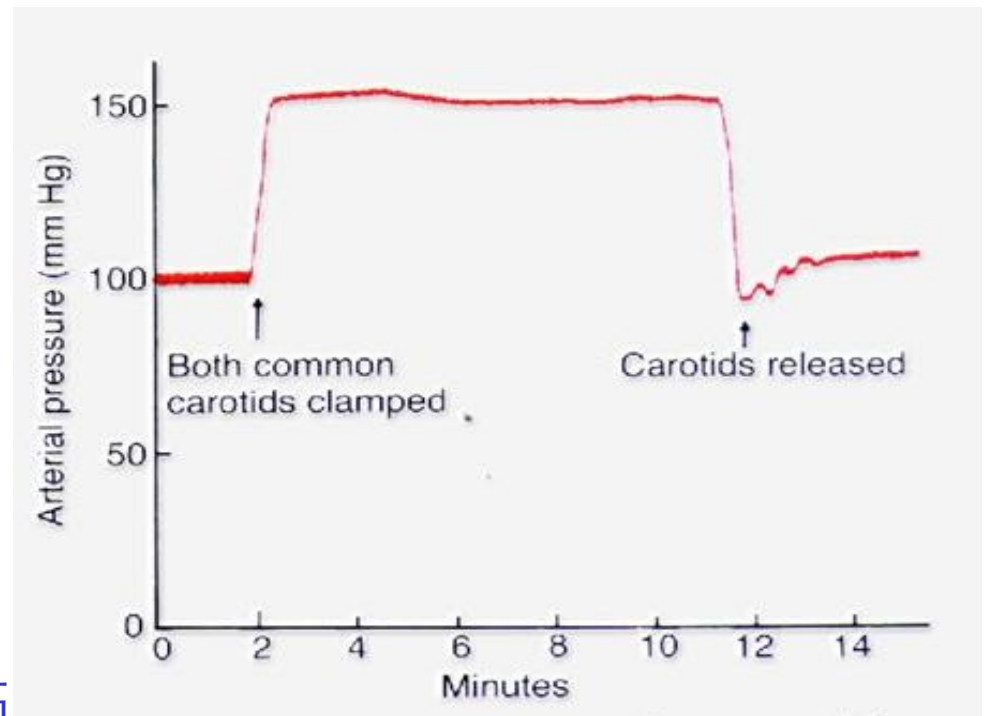
➤ 阻断颈总动脉血流（感受器传入冲动↓），血压反射性升高

实验：

➤ 再破坏迷走神经，血压更加升高
➤ 孤束核损伤，将引起神经性高血压

应用：

➤ 按摩颈动脉窦（感受器传入冲动↑），治疗心动过速。
➤ 儿童嬉闹触及颈动脉窦（感受器传入冲动↑），注意防止突然血压降低。
➤ 长期卧床者要定期摇床，注意防止反射废用。

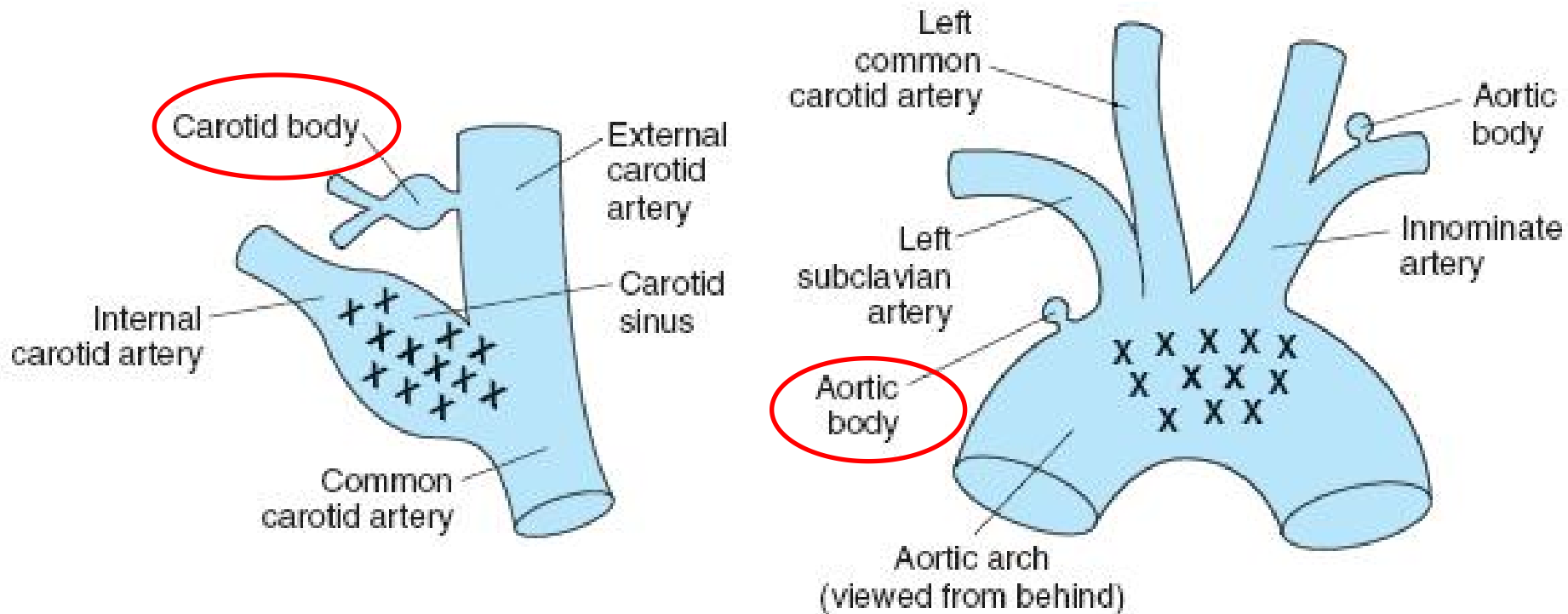


Chemoreceptor Reflex

化学感受器(chemoreceptor)

位置：颈动脉体和主动脉体

刺激：化学物质(P_{O_2} , P_{CO_2} , pH)





Corneille Heymans

(1892–1968)

- Corneille Heymans found baro-Rs, peripheral chemo-Rs. The latter in the control of respiration was awarded Nobel Prize in 1938.

C. Eyzaguirre S.J. Fidone
R.S. Fitzgerald S. Lahiri
D.M. McDonald
Editors

Arterial Chemoreception

With 106 Figures



1990年

Springer-Verlag
New York Berlin Heidelberg
London Paris Tokyo Hong Kong

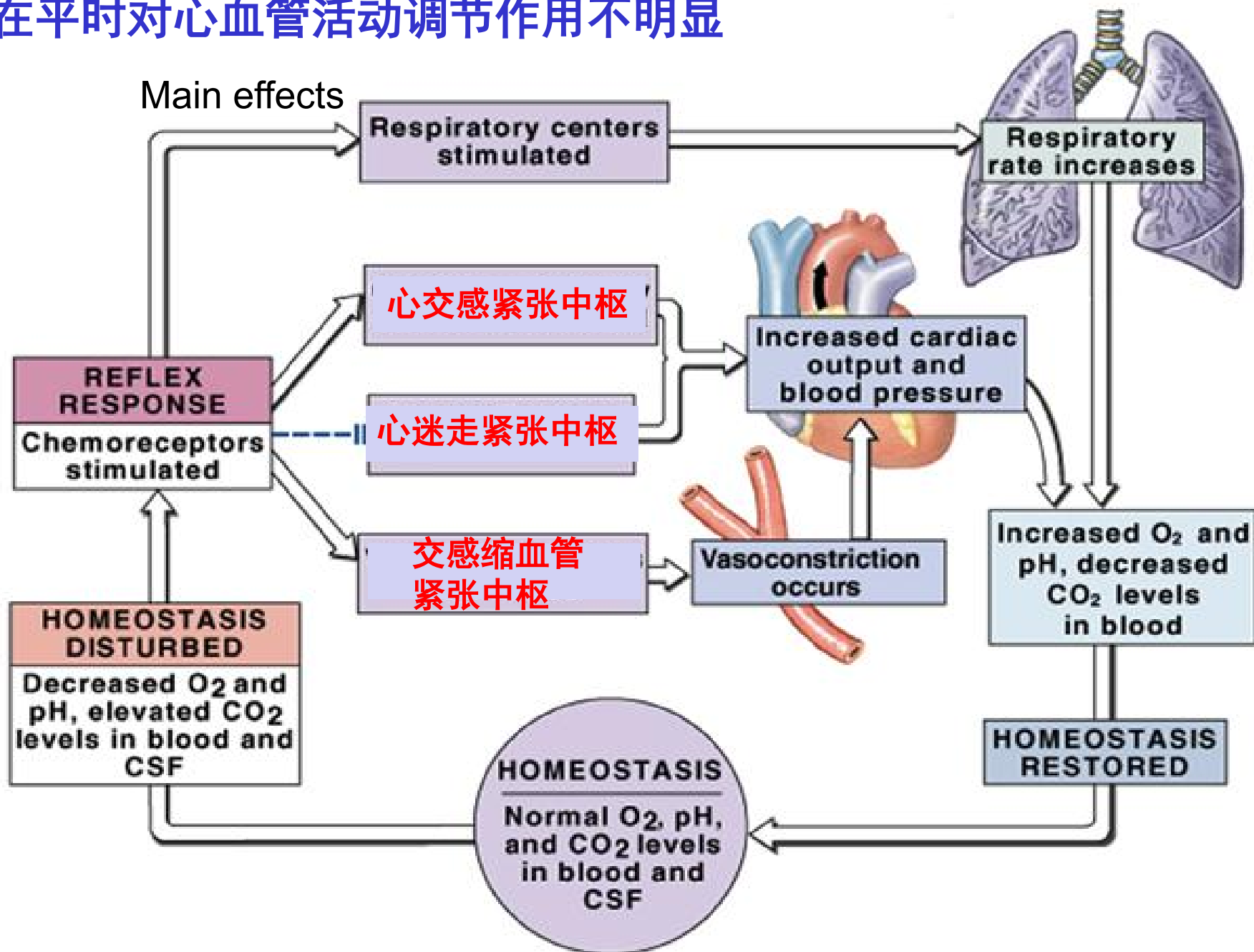
化学感受性反射的生理效应

- ✓ 保持自然呼吸的情况下，化学感受器的传入冲动可在引起呼吸加深加快的同时，出现心率加快，心输出量增多，外周阻力增大，血压升高等心血管活动的改变
- ✓ 人为保持呼吸的频率和深度不变，化学感受器的传入冲动引起心率减慢，心输出量减少，冠脉动脉舒张，骨骼肌和内脏血管收缩等效应。
(化学感受性反射兴奋迷走、兴奋交感，且迷走兴奋作用更强)

生理意义

- ✓ 主要调节呼吸运动，反射性的引起呼吸加深加快；通过呼吸运动的改变，再反射性影响心血管活动
- ✓ 在病理情况下兴奋交感缩血管中枢，使循环血量重新分配

在平时对心血管活动调节作用不明显

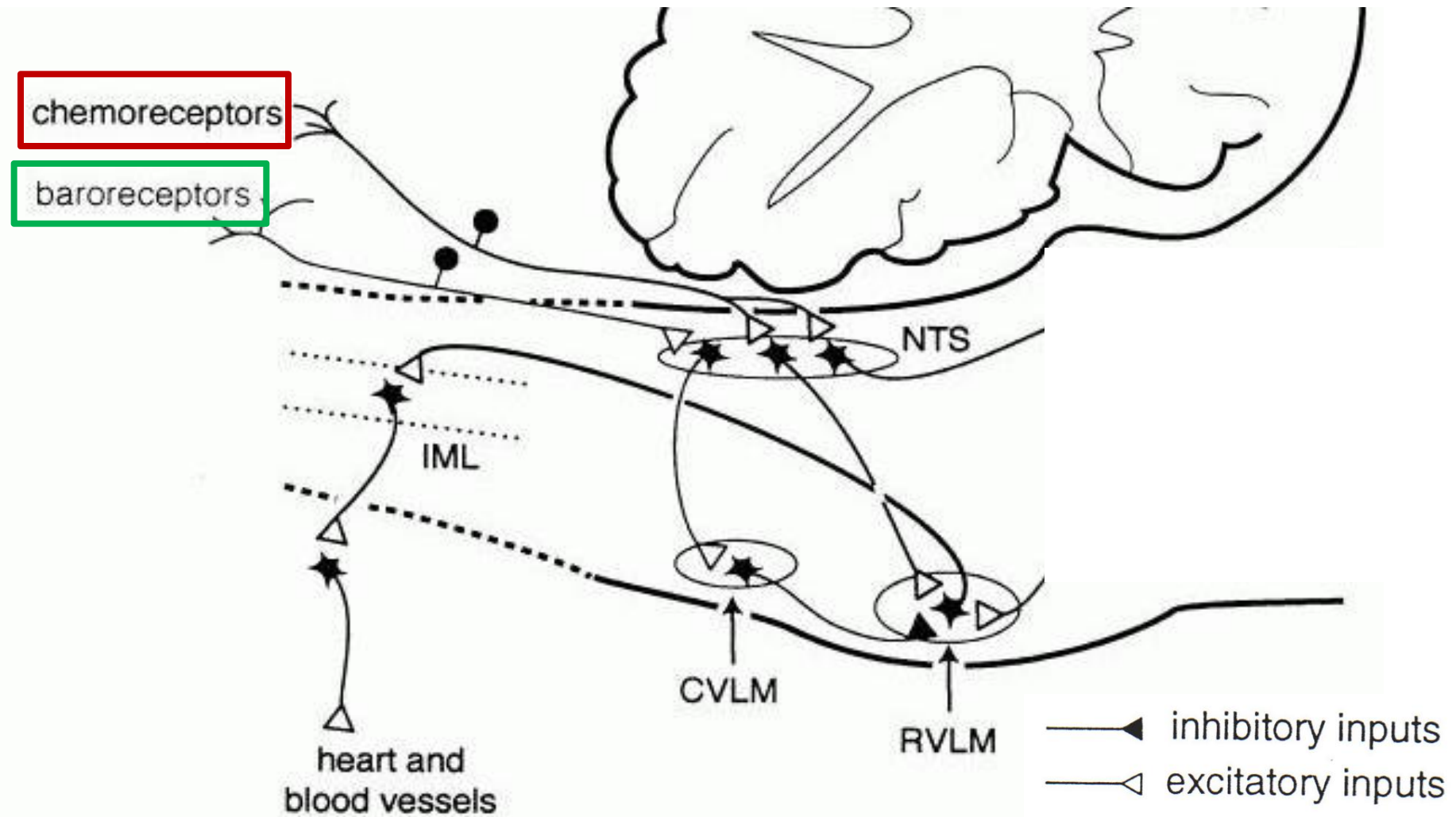


Inhibitory inputs

From carotid, aortic, and cardiopulmonary baroreceptors

Excitatory inputs

From carotid and aortic chemoreceptors



Cardiopulmonary receptors

心肺感受器：

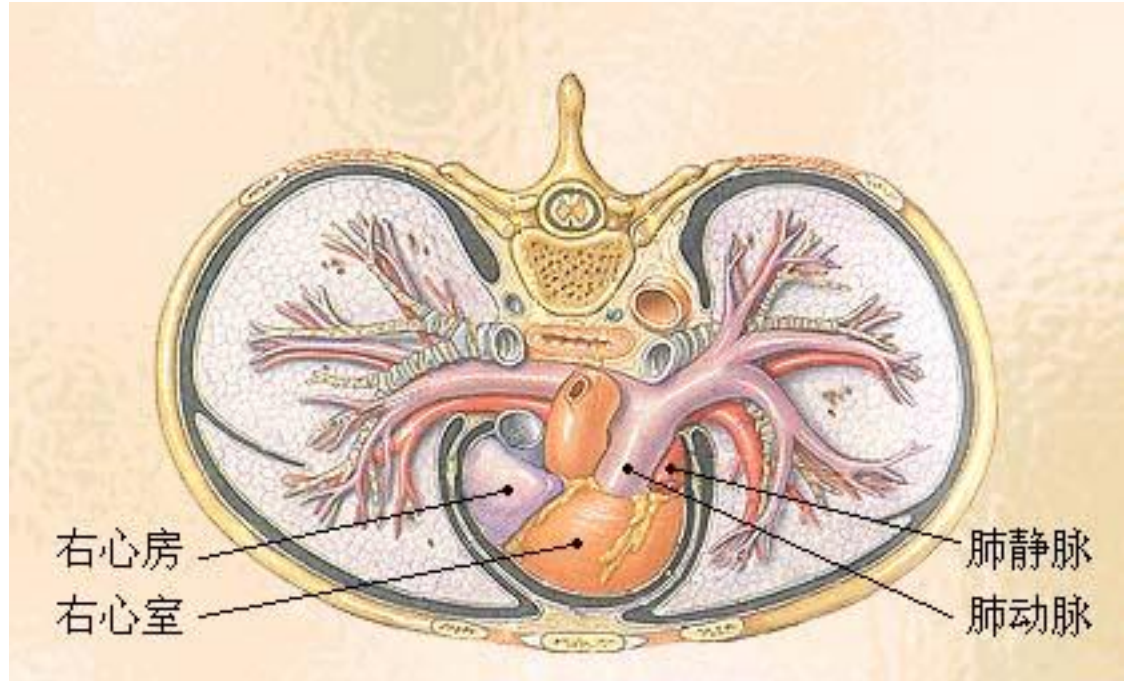
心房

心室

肺循环大血管壁

位于循环系统中

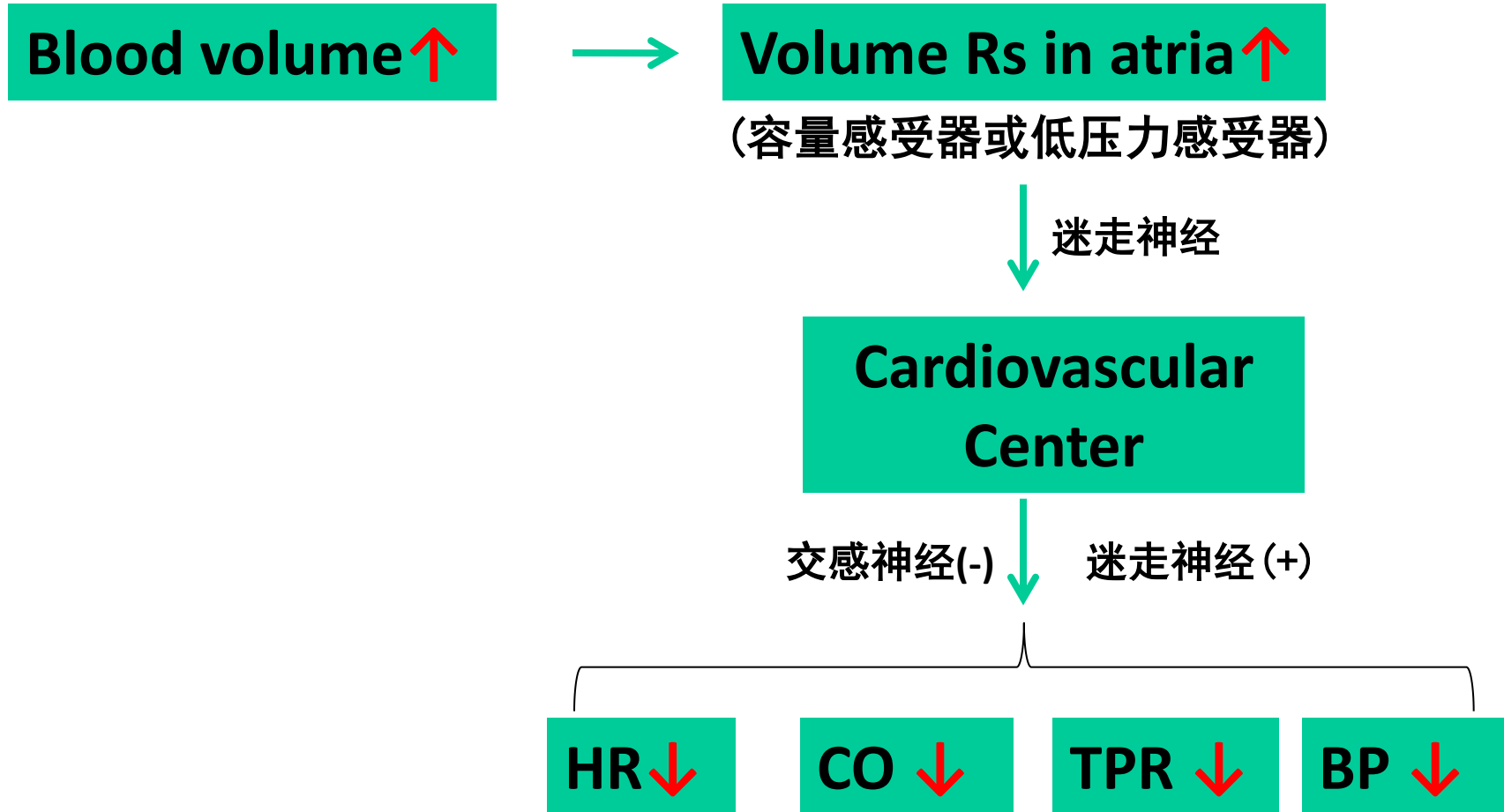
压力较低的部分。



感受机械牵张刺激 (+某些化学物质)

心房内的感受器又称为**低压力感受器/容量感受器**

Volume receptor reflex(容量感受性反射)

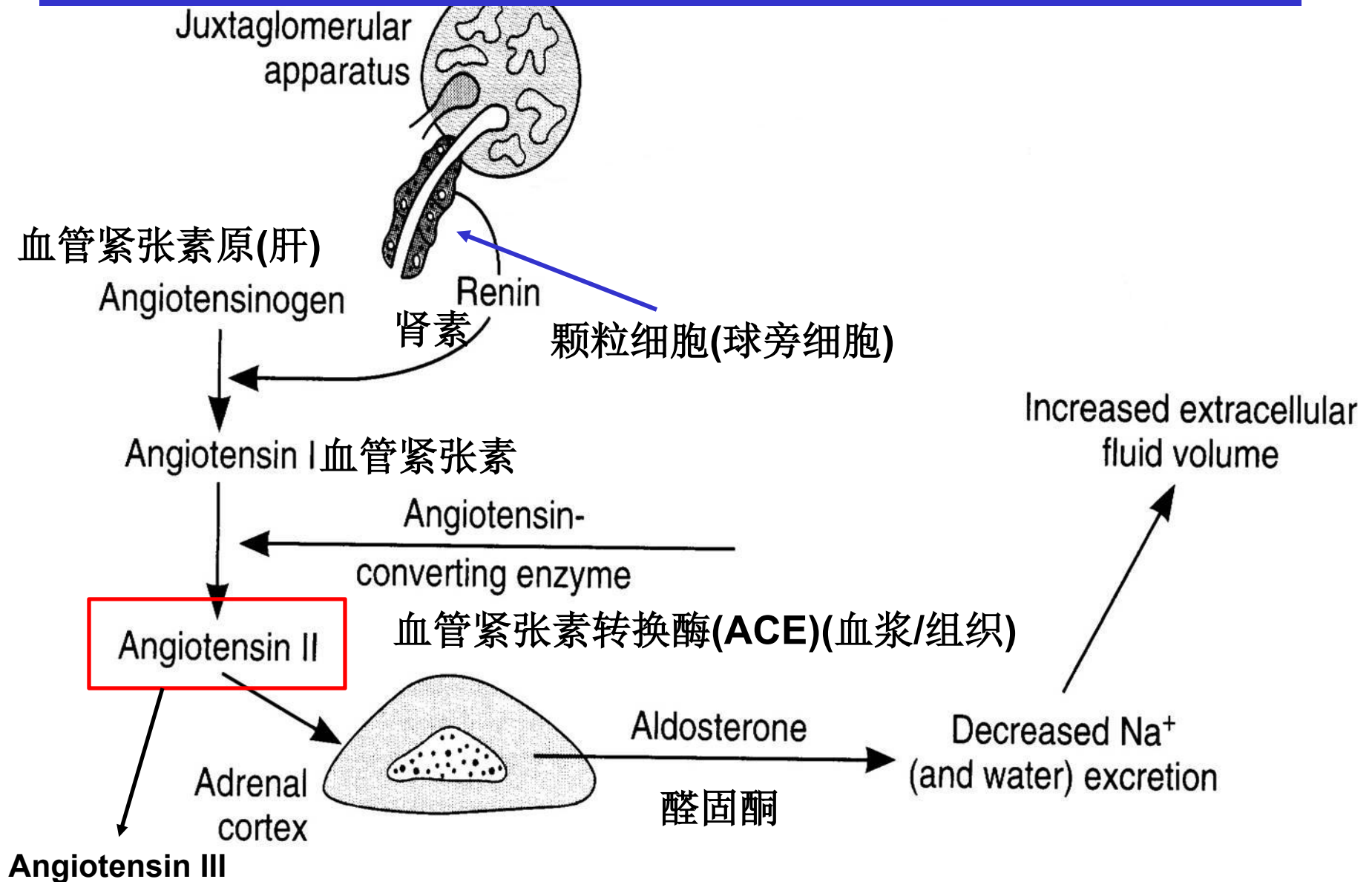


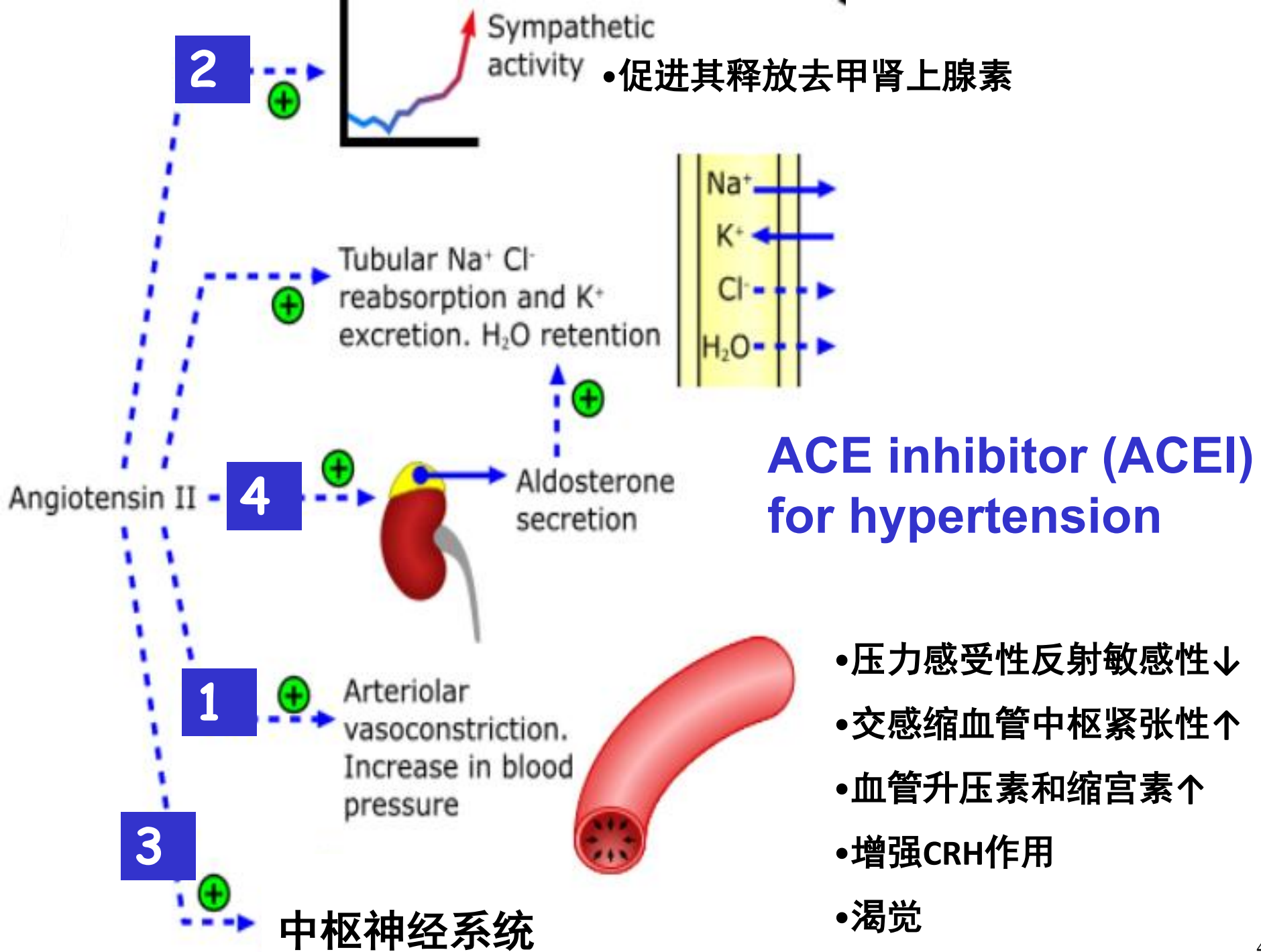
Hormonal Regulation

血液和组织液中化学物质对心肌和血管平滑肌活动的调节作用。

- Renin-angiotensin system
肾素-血管紧张素系统
- Epinephrine & Norepinephrine
肾上腺素和去甲肾上腺素
- Vasopressin (Antidiuretic Hormone-ADH)
血管升压素
- Atrial natriuretic peptide (ANP)
心房钠尿肽

• Renin-angiotensin system, RAS



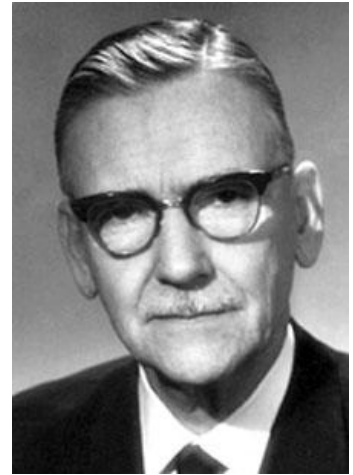


血管紧张素 I: 不具有生物活性

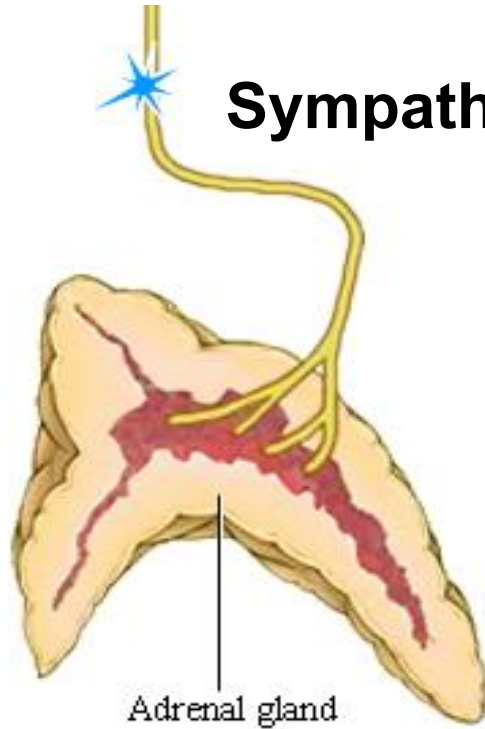
血管紧张素 III:

- **缩血管作用 < ANG II (10%-20%)**
- **促醛固酮合成释放作用 > ANG II**

•Epinephrine & Norepinephrine



•Ulf von Euler
(1970, Nobel laureates)
discovered NE & E



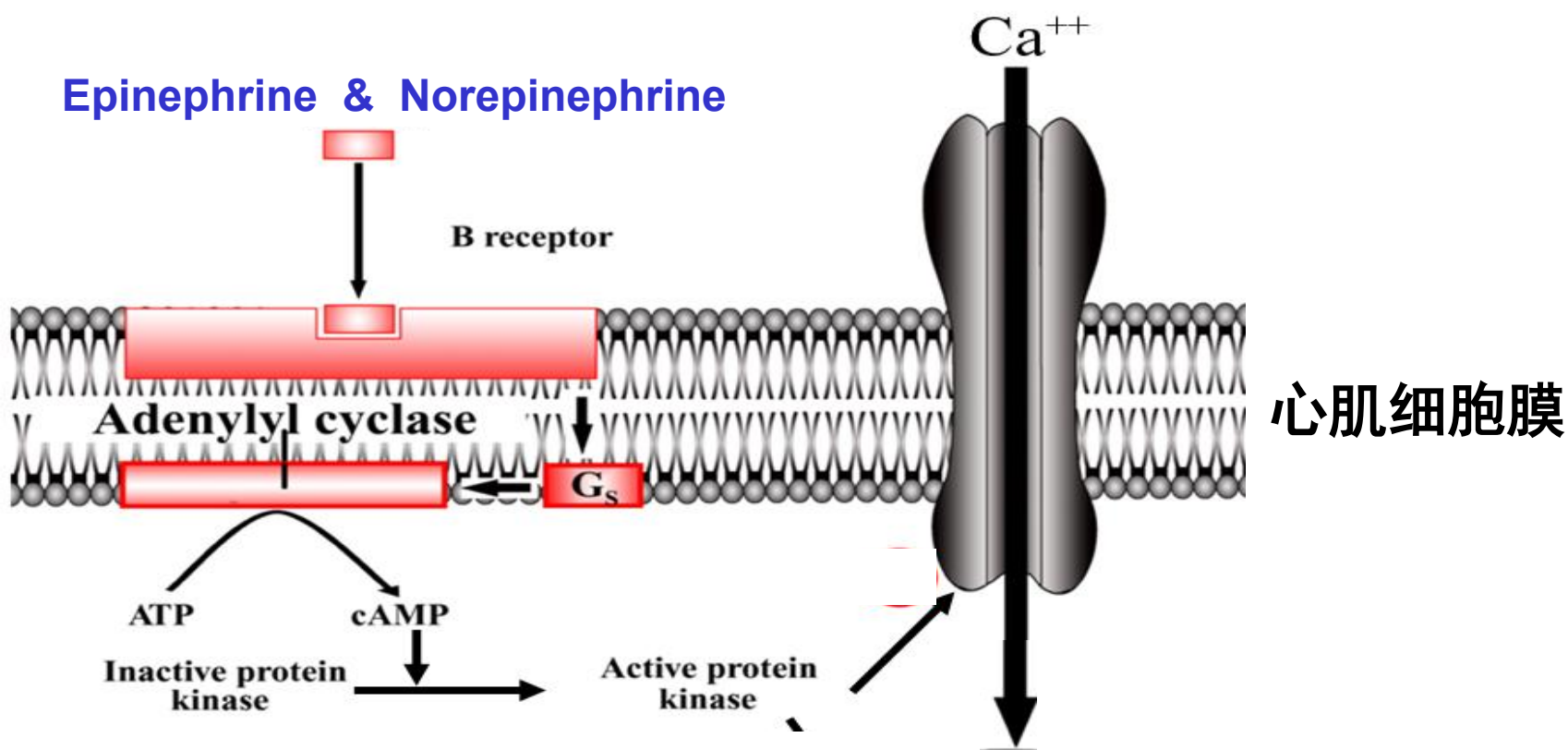
✓ 同属儿茶酚胺类

✓ 主要来自肾上腺髓质分泌

Epinephrine (E,80%)

Norepinephrine (NE, 20%)

✓ 循环中少量NE来自于交感神经末梢

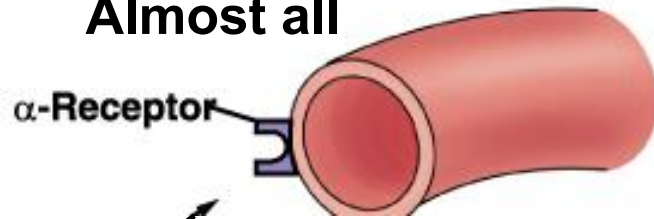


**E/NE — G蛋白—AC—cAMP↑-PKA
— Ca²⁺ 通道激活—Ca²⁺ 内流↑**

- **正性变时作用** — 心率增快
- **正性变传导作用** — 传导加快
- **正性变力作用** — 收缩增强

Almost all

α -Receptor



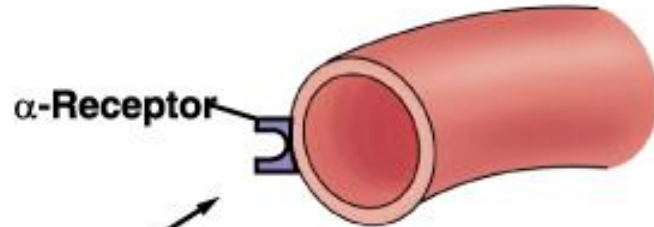
Norepinephrine

Low β affinity



Vessel constricts

α -Receptor

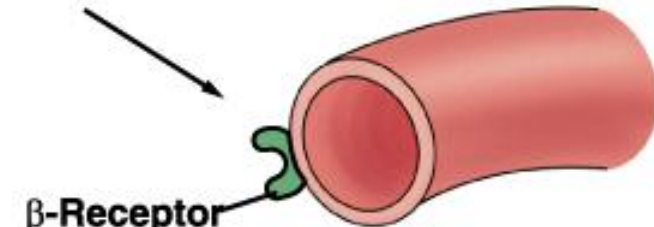


Vessel constricts

Blood vessels in skin, kidney, gastrointestinal...

Epinephrine

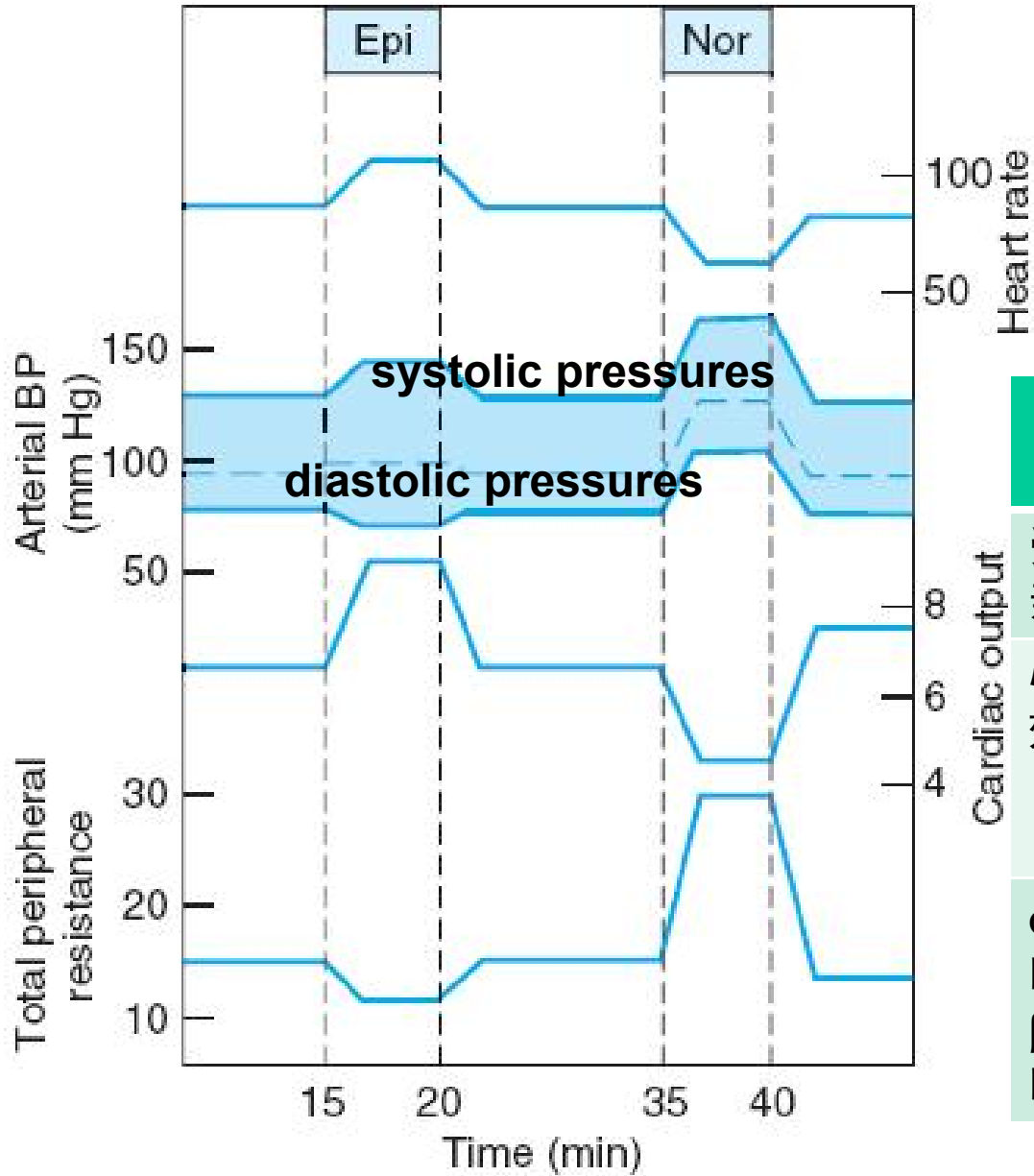
β -Receptor



Vessel dilates

Blood vessels in skeletal muscle, liver, heart...

•Circulatory changes produced in humans by the slow intravenous infusion of Epi and Nor.



	肾上腺素 E	去甲肾上腺素 NE/NA
受体亲和力	$\beta > \alpha$	$\alpha > \beta$
心脏效应	正性变时变力 心输出量↑	直接效应：同E 间接效应：心率↓
α 占优势的血管 β 占优势的血管	血管收缩 血管舒张	血管广泛收缩 外周阻力↑ 动脉血压↑

• Vasopressin (Antidiuretic Hormone-ADH)

血管升压素/抗利尿激素

来源

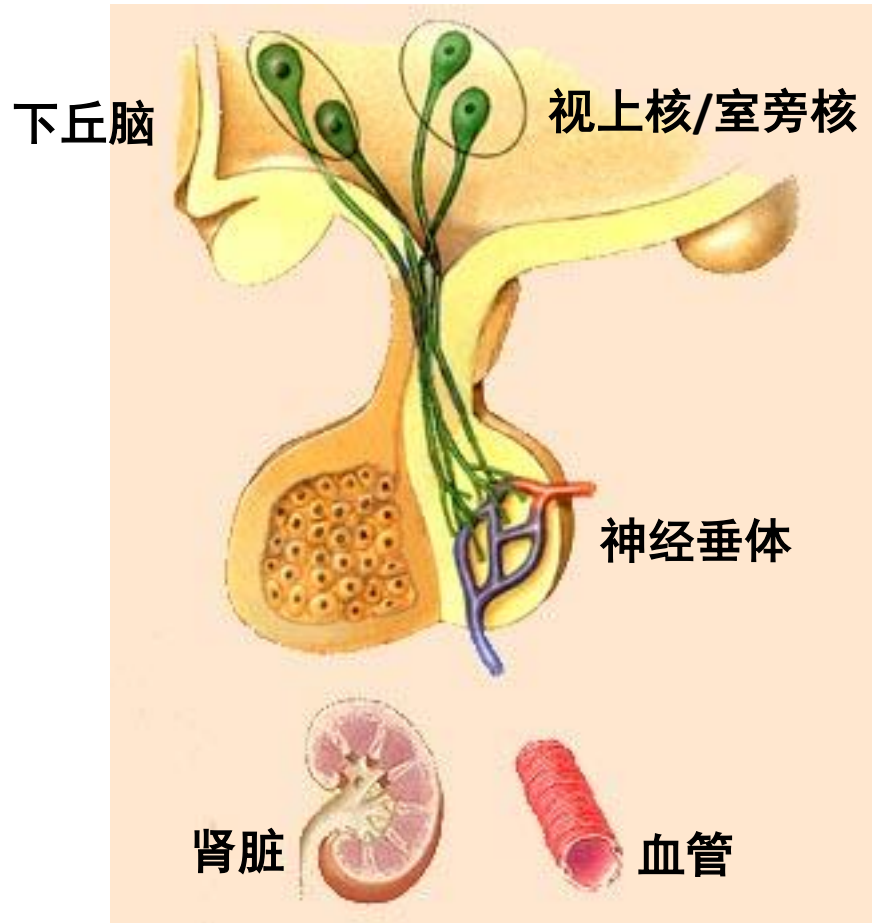
下丘脑视上核、室旁核

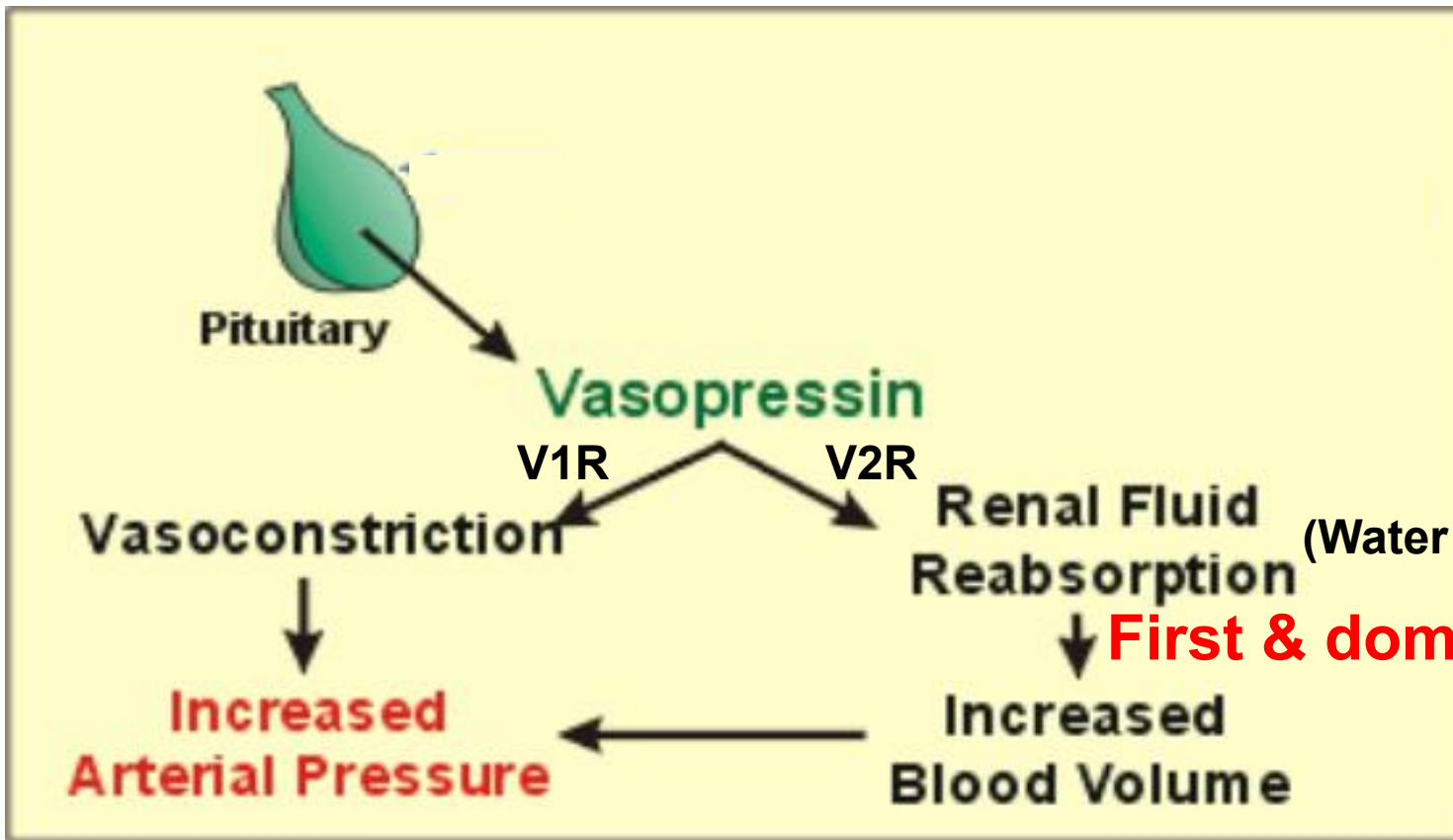


神经垂体贮存



释放入血





(Water Retention)

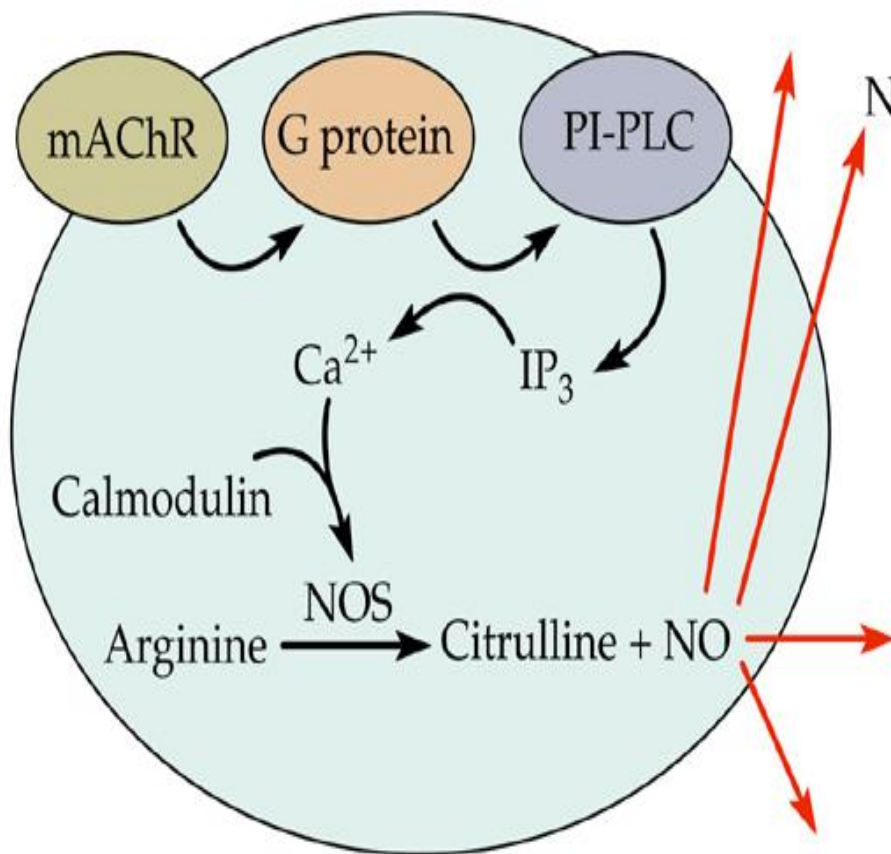
First & dominant effects

抗利尿效应

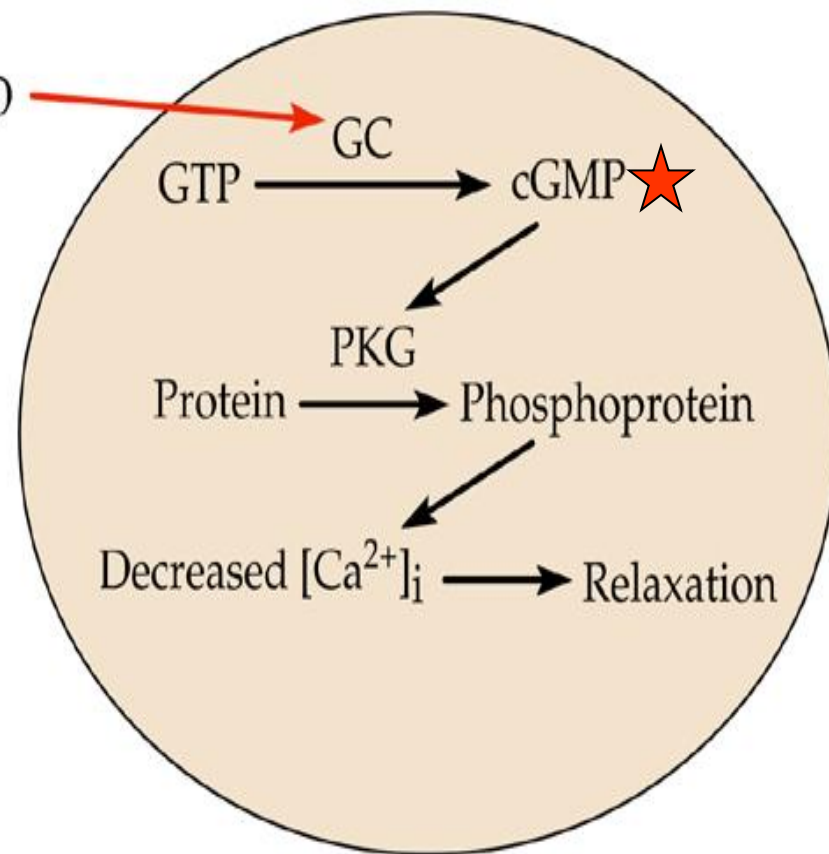
升高血压效应

Chemicals Mediating Vasoconstriction and Vasodilation

	来源	作用	临床价值
一氧化氮(nitric oxide, NO), 内皮舒张因子(EDRF)	Ach、缓激肽、切应力→内皮细胞→NOSIII*→L-精氨酸→NO→肌细胞→sGC*→GTP→cGMP→PKG→磷酸化MLCK→抑制MLCK→MLC磷酸化↓	舒张血管, 尤其是冠状动脉。与前列腺环素协同, 抑制血小板粘附、凝集	硝酸甘油治疗心绞痛、Viagra治疗阳痿。内源性NO减少导致高血压
内皮缩血管因子 内皮素, 22a.a. endothelin-1	血管内皮, 旁分泌作用于平滑肌	强烈缩血管, 旁分泌调节局部血管紧张性	
激肽释放酶-激肽 (kallikrein- kinins)	①血浆激肽释放酶→高分子激肽原→ 缓激肽 ; ②组织激肽释放酶(肾、唾液腺、汗腺、胰、胃肠)→高分子 / 低分子激肽原→ 缓激肽 / 赖氨酸缓激肽(血管舒张素)	通过NO舒张血管, 降血压, 类似组织胺增加毛细血管通透性。增加外分泌腺血流量。收缩内脏平滑肌	
前列腺环素 (Prostacyclin)	血管内皮细胞	舒张血管, 抑制血小板凝集	
5-羟色胺 (serotonin)	神经末梢, 血小板, 肥大细胞	缩血管, 调节局部血流, 与止血有关	
组织胺 (histamine)	神经末梢, 肥大细胞, 组织损伤、炎症时释放	舒张血管, 收缩内脏平滑肌	



Endothelial cell



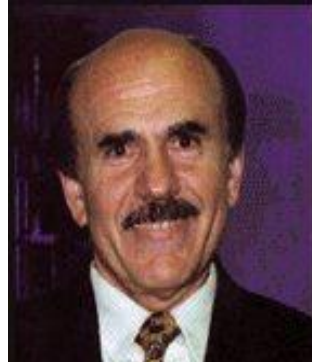
Smooth muscle cell

NO Signaling: 1998 Nobel Prize

- For their discoveries concerning **NO** as a signaling in cardiovascular system



R.F.Furchgot



L.J. Ignarro



F.Murad



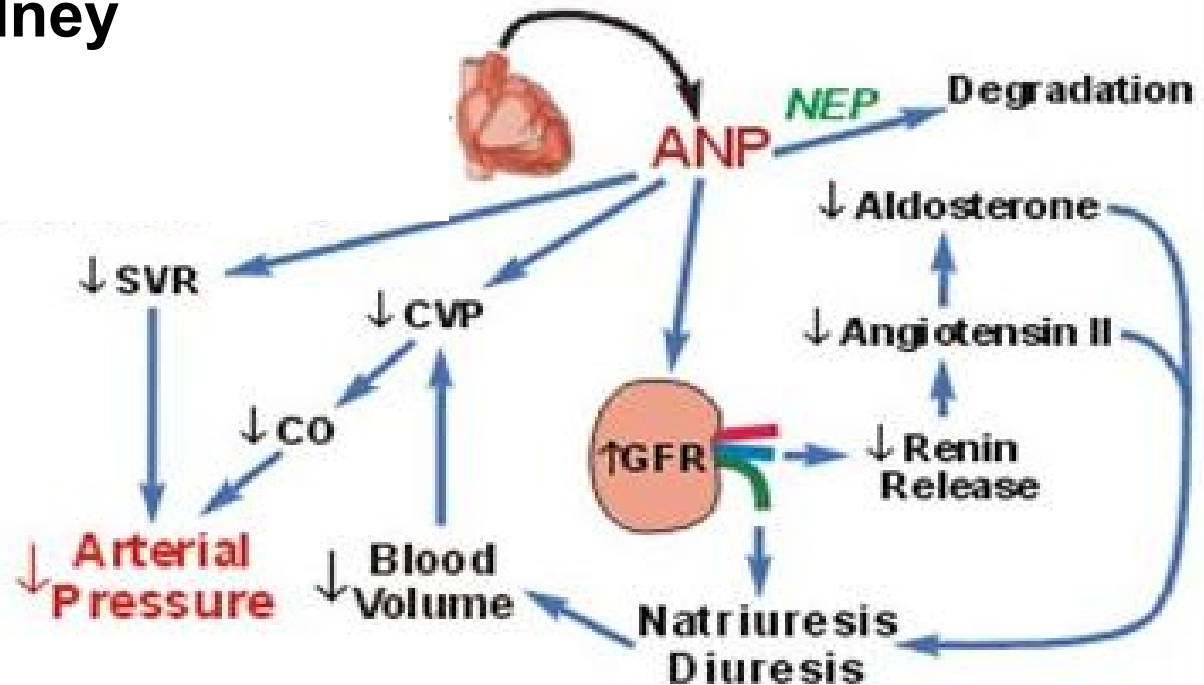
2006-9-21

• Atrial natriuretic peptide, ANP

Synthesized by atrium myocardium(心房肌)

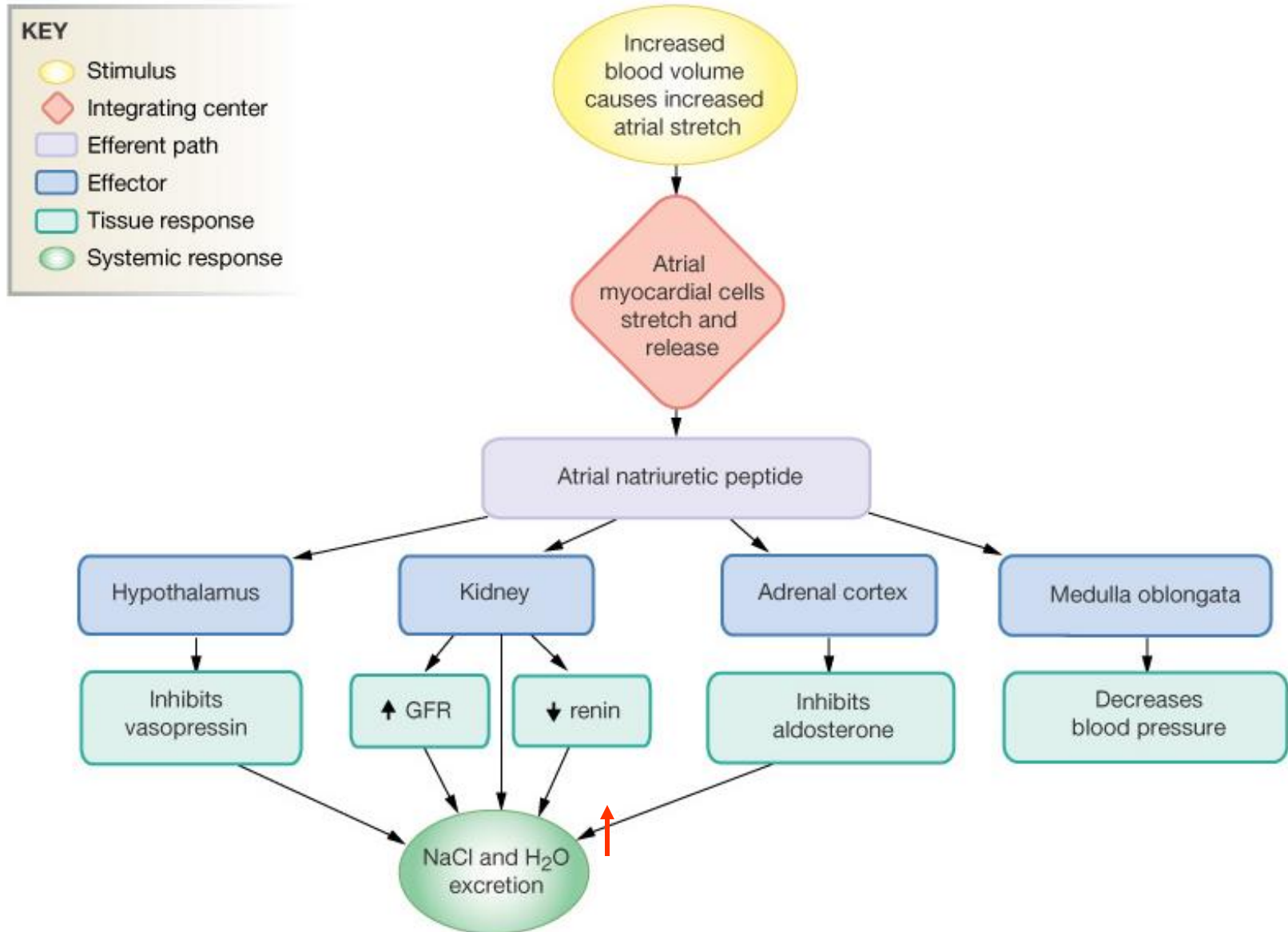
Main target - kidney

1. 利钠利尿作用
2. 心血管作用：
舒张血管、抑制
心脏活动



Abbreviations: ANP, atrial natriuretic peptide; NEP, neutral endopeptidase; GFR, glomerular filtration rate; CVP, central venous pressure; CO, cardiac output; SVR, systemic vascular resistance.

Atrial Natriuretic Peptide: Regulates Na^+ & H_2O Excretion



Autoregulation

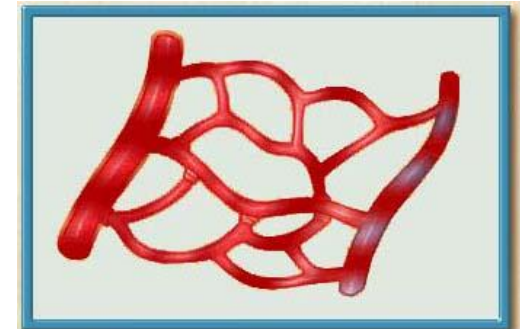
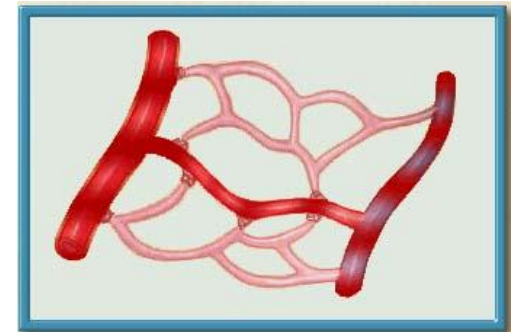
The capacity of **tissues** to regulate their own blood flow is referred as autoregulation

1. 代谢性机制

✓ 后微动脉和毛细血管前括约肌交替开放

- $P_{O_2} \downarrow$ 或者代谢产物 $\uparrow$ 引起舒张

CO_2 、腺苷、乳酸、 H^+ 、 K^+



2. 肌源性机制（尤其是毛细血管前阻力血管）

Small arteries and arterioles response to perfusion pressure

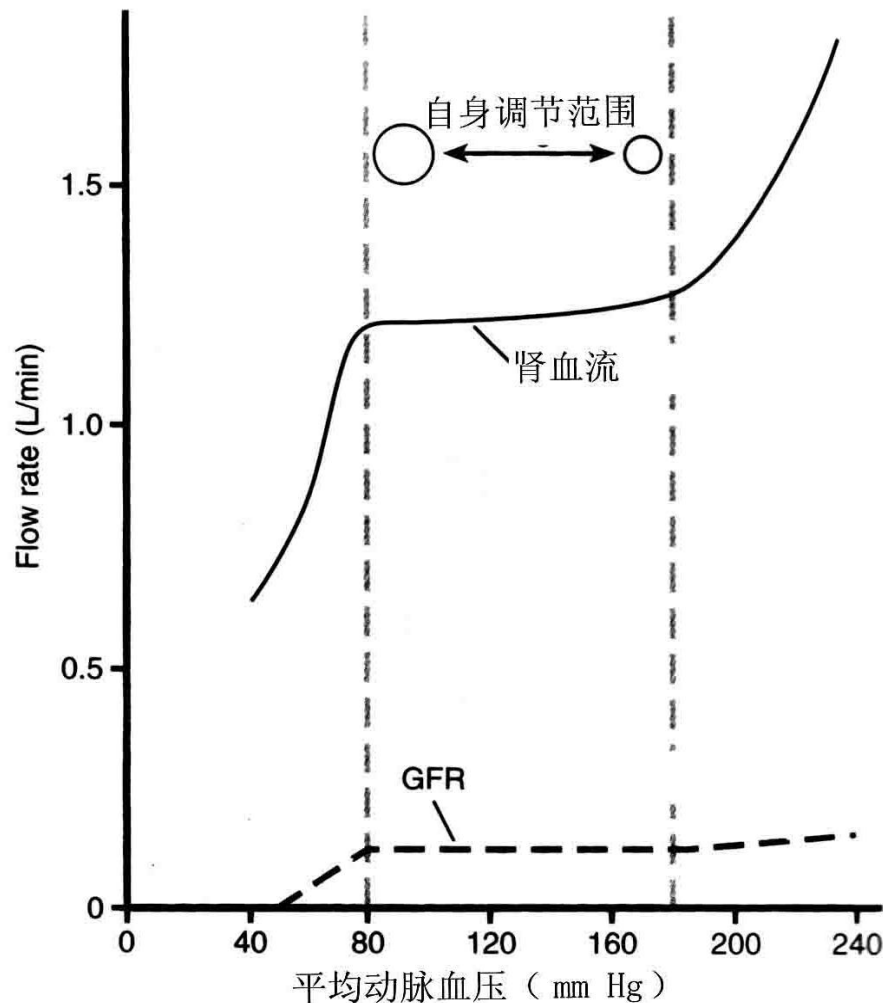
灌注压↑

血管平滑肌收缩

血流阻力↑

血流量↓

✓在肾血管中特别明显



Long- Term Regulation of AP

- On a time scale of hours or days-
Regulation occurs via pathways that target **kidneys**, in their control of extracellular fluids (ECF).
- **Renal-body fluid system (肾-体液控制系统)**

4. 躯体感受器引起的心血管反射

刺激躯体传入神经(刺激的强度和频率)→心血管反射

5. 内脏感受器引起的心血管反射

扩张内脏器官→心血管反射

压迫眼球、压迫腹部→心血管反射

6. 脑缺血反应(brain ischemia response)

脑血流↓→心血管反射